

LCN System Checkout

SW20-610

LCN Installation

LCN System Checkout

SW20-610

Release 610

CE Compliant

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About This Publication

This publication provides instructions for use by the system hardware installation and checkout personnel. It will help you determine if the system hardware is properly installed and ready for on-line operation. Use this manual whenever the system is shut down for major repairs, upgrades (expansions) or major moves (relocation of equipment).

The user of this manual should be familiar with the contents of the *TPS System Site Planning* manual and *LCN Planning* manual and the *LCN System Installation* manual.

Global User Station (GUS) is part of Honeywell's **TotalPlant** Solution (TPS) System. TPS is the evolution of TDC 3000^X.

This publication supports TPS software release 500, 600 and CE Compliant hardware.

Any equipment designated as "CE Compliant" complies with the European Union EMC and Health and Safety Directives. All equipment shipping into European Union countries after January 1, 1996 require this type of compliance—denoted by the "CE Mark."

The Global User Station tests in the Core Module Test System (CMTS) have been added.

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1. INTRODUCTION

Section 1

This section lists all support documents and services available from Honeywell. It also lists tools and test equipment that may be needed during system checkout.

1.1 PURPOSE OF THIS MANUAL

This System Checkout Guide provides instructions and references, as needed, for initial off-process checkout of TPS equipment cabinets and Universal Stations/Universal Station^X stations containing modules, gateways, monitors, and peripherals. It is intended for use by trained Honeywell or customer service technicians. The checkout procedures will determine if the system components have been properly installed and are ready for system startup. This guide also serves to check out system expansions or significant modifications to existing systems.

For checkout of TPS Data Hiway BASIC equipment, Operator Stations, controllers, Process Interface Units, and related products, reference is made to applicable Data Hiway equipment checkout and service manuals.

The TPS System Post-Installation Checklist, found in Appendix D, must be used by the responsible technician to measure the progress and completeness of the checkout effort.

A selected list of operating practices and general housekeeping recommendations is found in Appendix A. Each item on the list should be acknowledged by the responsible party and action taken before and during system startup.

Read this guide thoroughly and scan the following referenced documents in section 1.2 before proceeding with system checkout.

1.2 HONEYWELL SUPPORT SERVICES AND DOCUMENTS

Optional Power-on support, Field Services, and Technical Support are available during the on-site checkout of the TPS System equipment. The following Honeywell support documents are required or are of assistance during checkout and operation of the system.

Title	Binder
TPS System Site Planning	System Site Planning - 1
LCN Planning	System Site Planning - 1
LCN System Installation	LCN Installation
LCN Guidelines - Implementation, Troubleshooting, and Service	LCN Installation
System Startup Guide, Cartridge Drive	Implementation/Startup & Reconfiguration - 1
System Startup Guide, Floppy Drives	Implementation/Startup & Reconfiguration - 1
Network Form Instructions	Implementation/Startup & Reconfiguration - 1
Configuration Data Collection Guide	Implementation/Startup & Reconfiguration - 2
Process Operations Manual	Operator/Process Operations
History Module Service	LCN Service - 2
Five/Ten-Slot Module Service	LCN Service - 2
Dual Node Module Service	LCN Service - 2
Maintenance Test Operations	LCN Service - 1
Universal Station Service	LCN Service - 1
Universal Station ^X Service	Universal Station ^X
Universal Station (Ergonomic) Service	LCN Service - 1
Universal Station ^X (Ergonomic) Service	Universal Station ^X
Deskside Global User Station Service	Global User Station
Console Global User Station Service	Global User Station
Classic Console Global User Station Service	Global User Station
Hardware Verification Test System	LCN Service – 3
LCNI Network Communications Test	LCN Service - 3
Test System Executive	LCN Service - 3

Title	Binder
Core Module Test System	LCN Service - 3
Universal Control Network Installation	Installation/Universal Control Network
Process Manager Installation	Implementation/Process Manager - 1
Process Manager Checkout	Implementation/Process Manager - 1
Advanced Process Manager Installation	Implementation/Advanced Process Manager - 1
Advanced Process Manager Checkout	Implementation/Advanced Process Manager - 1
Process Manager/Advanced Process Manager Service	PM/APM/HPM Service-1
Process Manager Module Test System (PMMTS)	PM/APM/HPM Service-2
Process Manager Test System (PMTS)	PM/APM/HPM Service-2
Process Manager Test Executive (PMEX)	PM/APM/HPM Service-2
PLC Gateway Control Functions	Implementation/PLC Gateway
PLC Gateway Parameter Reference Dictionary	Implementation/PLC Gateway
PLC Gateway Forms	Implementation/PLC Gateway
Network Gateway Site Planning and Installation	Implementation/Network Gateway
Basic System Checkout	Basic System Summary
Basic System Cabling and Wiring	Basic System Summary
Data Hiway Subsystem Site Planning	Basic System Summary
PIU Installation	Basic System Summary

1.3 TOOLS AND TEST EQUIPMENT

- Test Operator Control Station (that is, Universal Station) with cartridges or floppies
- RS-232C-compatible I/O device (TI700)
- I/O Paddle Board 51107954-100
- Test Diskettes/Cartridges (see checklist)
- Modem cable kit
- CLI Test cable 51109040-100
- CLI Test connectors 51109041-100 (two required)
- Digital Voltmeter
- Standard tool kit
- ESD ground-strap
- Extra LCN T-connectors
- Time Domain Reflectometer, TDR, Cable Tester, Tektronix 1502*, range 0 to 2000 feet, 6 inch resolution
- Time Domain Reflectometer, TDR, Cable Tester, Tektronix 1503*, range 0 to 50,000 feet, 3 foot resolution, plus:

Selected 75 Ohm terminator (must be 75.0 + or - 0.1 ohms, that is, 74.9 to 75.1 ohms) (select from C-KC02 or 30732052-001)

- Time Domain Reflectometer, TDR, Cable Tester, Riser Bond 1270A*
- BNC, jack-to-jack adapter or barrel splice (C-KCA04 or 30732051-001)
- Voltage Analyzer, Dranetz series 626*
- Biddle Megger® earth-ground tester, 250200 or 250241*
- Distortion analyzer - HP 334*
- Oscilloscope - Tektronix 465 or equivalent*
- Power Analyzer, Dranetz-BMI PowerVisa 440 **
- Fluke 1625 Kit or AEMC 3731 Clamp-on Ground Resistance Tester ***

*System test devices must be available as needed.

**New test devices. Dranetz-BMI PowerVisa 440 can be used instead of Dranetz Series 626 Analyzer, and it does not need dc plug-in modules.

***New test devices. AEMC 3731 Clamp-on Ground Resistance Tester or equivalent can be used in place of the Biddle Earth Ground Tester.

2. SYSTEM EQUIPMENT CHECKOUT

Section 2

This section lists the major hardware items to be checked and what to look for.

2.1 CHECKOUT OVERVIEW

The *System Checkout Guide* must be referenced during and after system installation. This is to ensure that the equipment is properly installed, intercabled, and powered up. Final checkout is performed by loading and running the Hardware Verification Test System (HVTS), the Core Module Test System (CMTS), and the LCNI Network Test diskettes/cartridges from the Universal Station. Test procedures are provided in the HVTS publication listed in subsection 1.2. See 3.1 for HVTS use-scenario and 3.2 for BASIC System equipment startup.

For system startup instructions and floppy diskettes/cartridge disks required, refer to the *System Startup Guide* in the *Implementation/Startup & Reconfiguration - 1* binder.

This manual, along with those listed in subsection 1.2 and the test diskettes/cartridges, provides all the information necessary to verify proper installation. It also ensures that the system is ready for configuration and startup.

2.2 POWER AND GROUNDING

Check existence and adequacy of all system-power requirements, as described in the *TPS Site Planning Manual* or the *Data Hiway Subsystem Site Planning Manual*, and the *Data Hiway Cabling and Wiring Guide*. Check integrity of ac safety ground, logic ground, and Data Hiway ground.

2.2.1 AC Safety Ground

Visually inspect each cabinet and console to verify existence and proper connection of the ac safety-ground wire. A frame-ground service post is provided at the bottom of each cabinet and console bay for this purpose only (see Figure 2-1 and Figure 2-2.) Ensure that only the safety-ground wire from the system power-distribution panel is connected to the frame-ground service post.

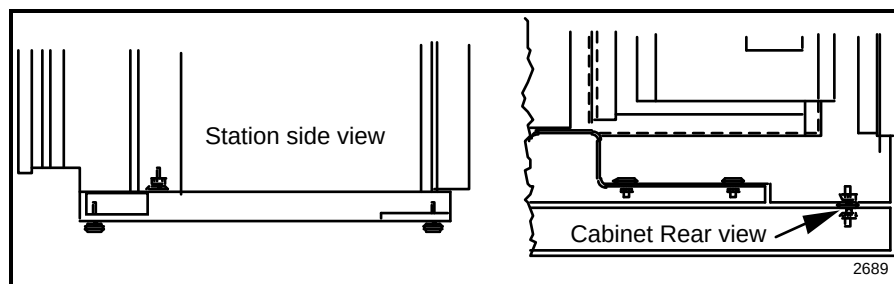


Figure 2-1 — AC Safety Ground Service Post Location

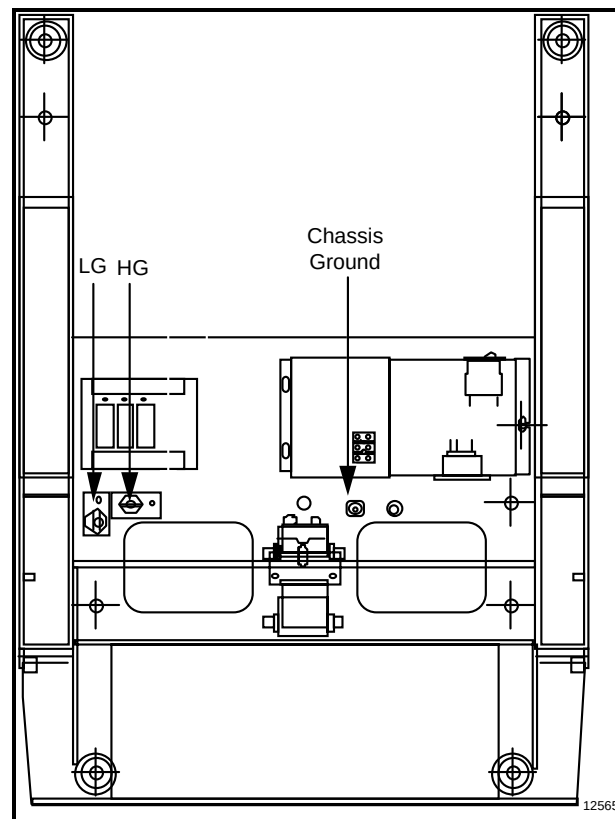


Figure 2-2 — AC Safety Ground Service Post Location in Ergonomic Furniture

2.2.2 Data Hiway Ground

If the system includes a Hiway Traffic Director (HTD), the Data Hiway cable-shield is grounded only at the HTD, as illustrated in Section 5 of the *Data Hiway Subsystem Site Planning Manual*. Also, the wires on HG terminals 11 and 13 must be disconnected and separately taped-up (with insulated electrical tape) to prevent cable shields from shorting together.

If the system does not include the HTD, the hiway cable-shield must be grounded at the primary Hiway Gateway module, as follows (also illustrated in Section 5 of *Data Hiway Subsystem Site Planning*):

- At TB1 on the Gateway I/O DHI paddle-board, ensure that terminal screws 11 and 13 are wired (with #14 AWG) to the Data Hiway (DH) service-post at the bottom of the cabinet or console. The DH service-post, must be connected to Master Reference Ground (MRG) through a #4 AWG wire. Note that the HG service post is also insulated from the cabinet/console frame because it is mounted on the same phenolic block as the LG service post.
- Make sure that the Data Hiway Ground is connected to only the Master Reference Ground rod.

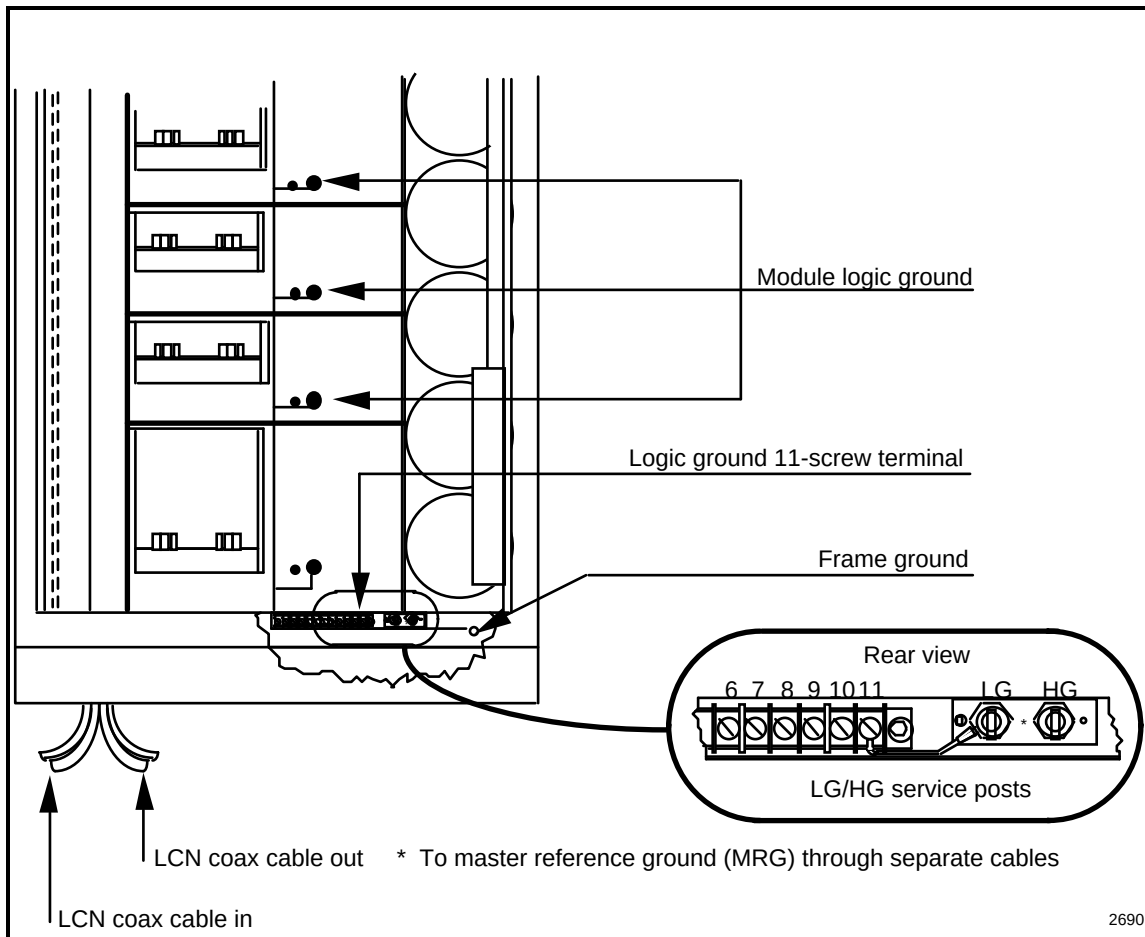


Figure 2-3 — Cabinet Logic Ground Terminal Board

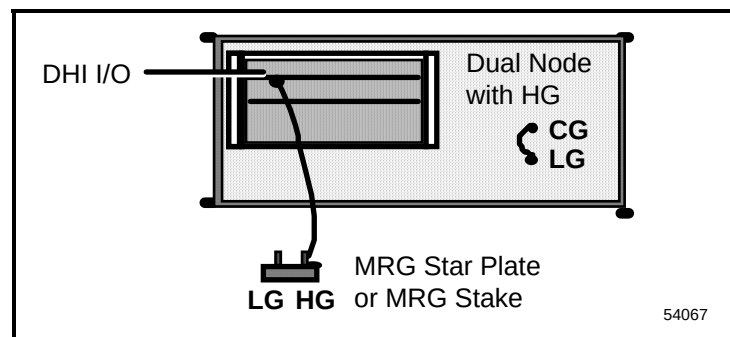


Figure 2-4 — Logic Ground Connection for Five/Ten-Slot Modules

2.3 LCN/HIWAY CABLING

If the system includes both Hiway Gateways (HG) and Basic or Enhanced Operator Station(s), with or without a Hiway Traffic director (HTD), check all preferred-access cabling for proper termination and grounding. Installation rules are presented in Section 3 of the *Data Hiway Subsystem Site Planning Manual*.

Check the following:

- Check LCN cabling for proper use of T-connectors and terminators.
- Ensure that all cables labeled "Cable A" are connected to the "A" terminal.

T-connectors on the I/O LCN paddle-board (P/N 51107403-100). This board is always located in slot 2 of the small-card cage or bustle of each LCN module (except in the case of Dual Node Modules).

- Ensure that all cables labeled "Cable B" are connected to the "B" terminal only.

T-connectors of each module.

- After power-up (subsection 2.5), note that the TX (amber) indicator on the large LCN board in slot 2 of each module glows, indicating that the token ring is established.
- Check all Data Hiway boxes for proper use of T-connectors and terminators, and for proper routing of Hiway cables A and B.
- Conduct Data Hiway cable Time Domain Reflectometer (TDR) tests, as required, before burial or before it is placed in service. TDR test procedures for the Data Hiway cables are found in the *BASIC System Checkout Guide*.
- Depending on the TDR equipment available, use TDR test procedures for LCN cable segments as follows:
 - Appendix C: TDR test procedures for LCN cable segments using Tektronix TDR model 1502/1503.
 - Appendix E: TDR test procedures for LCN cable segments using Riser Bond TDR model 1270A.

2.4 XLCNE2 CABLING

If the system includes the Local Control Network Extender (XLCNE2), make sure that the cable "A" subsystem is the "Cable" subsystem to all nodes. Check the cable "B" subsystem for the same reason; there must be no crossing of cables. The cable subsystem includes all coaxial cables bus segments, fiber optic cables, and XLCNE2s. Check for the following application requirements/limitations:

- Up to three coaxial cable bus segments can be serially connected through XLCNE2s and fiber optics cables. The maximum length of the fiber optics cables (that transmit and receive) is 2 km, 6562 feet.
- The XLCNE2 circuit board can be installed in any LCN module, in any spare paddle-board slot.
- The XLCNE2 does not have an LCN (node) address; however, it counts as one of the 40 allowable loads on a coax-cable bus segment.
- The real-time clock data signal is not repeated by the XLCNE2. Therefore, a Clock Source/Repeater (CS/R) paddle-board is required on each LCN coax-cable bus segment (one for Cable A and one for Cable B). The CS/R is installed in the paddle-board slot behind the EMPU/HMPU/HPK2 circuit board. There are two different methods of providing clock communications, where (1) the remote bus segment is under 300 meters (984 feet), using shielded-twisted pair, (2) the remote bus segment is up to 2 kilometers (1.2 miles) from the main coax segment, using fiber optic cable. Refer to the *LCN Guidelines - Implementation, Troubleshooting, and Service* manual in this binder.

If a remote segment has only K2LCN boards in the modules, the digital clock (5 MHz) is used and no special transmission link is needed for the clock.

2.4.1 Operational Checkout of LCN Extenders (XLCNE2s)

Each fiber optic interface board (XLCNE2, FOCR) has a green LED indicator, which indicates that the minimum acceptable light power level is received. The yellow indicator indicates that the data is passed by the receiving board. These indicators can be used for establishing that the system is operational.

1. With both the modules (transmitting and receiving) powered on and the fiber optic cable connected, the green LED must be in the status "on". If not, refer to the *LCN Guidelines - Implementation, Troubleshooting, and Service* in this binder for troubleshooting procedures.
2. With both modules in the status "on" and token-passing taking place (the yellow LED on the LCNI boards blinking), the yellow LED on the receiving boards (XLCNE2, FOCR) must blink. If this does not occur, refer to the *LCN Guidelines - Implementation, Troubleshooting, and Service* in this binder for troubleshooting procedures.
3. Perform power loss measurements as described in *LCN Guidelines - Implementation, Troubleshooting, and Service* in this binder.

2.5 MODULE/BOX SELF-TEST

All the Local Control Network (LCN) modules, HPN modules, and almost all the Data Hiway (DH) boxes perform self-test upon power-up (system/unit power switch on). This requires a visual check of the LED-display status on each module/box in the system. Remove module covers to observe all LED indicators. If a failure is detected, the box or module in question detects a hardware or firmware failure and must be serviced or replaced before running any software tests.

For the Module LED status-test-display meaning and corrective action, refer to the *Five/Ten-Slot Module Service* manual, in the *Service* binder.

If a module or box failure occurs later, when the system is on line, the operator maintenance personality logs the failure and recommends corrective action at the Optimum Replaceable Unit (ORU) level.

For the Hiway Box LED-status displays, their interpretation and corrective action, refer to the *BASIC System Service Manual*, *TDC 355 Troubleshooting Flow charts*.

2.6 COLOR MONITOR CHECK

The color monitor checkout consists of a visual inspection of the power cord and color cables to ensure that they are properly installed. To check out the color monitor, perform the following steps:

1. After the Universal Station is powered UP, press the Degauss button on the right of the screen.
2. Hold the button for a second or two to ensure that colors are restored.
3. Adjust the Brightness and Contrast thumb wheels on the right of the screen. This is for the required effect on the blinking character that appears on the upper-left of the screen.

NOTE

If the brightness and contrast adjustments are too high, and a pattern remains on the screen too long, the screen phosphor may be etched, or worse yet, permanently burnt. The etch pattern can be removed by flooding the screen with white for an extended period of time.

For service information, including adjustments, test, and troubleshooting, refer to sections 3 through 6 of the *Universal Station Service* manual, in the *Service* binder.

2.7 PRINTER SELF-TEST

2.7.1 ASPI-41 Printer

1. Ensure that there is ribbon and paper in the printer and that the top cover is closed.
2. Press and hold the **ON LINE** button while setting the **POWER** switch to **ON**, then release the **ON LINE** button.
3. The printer display first shows **ASPI**, then the printer begins its self-test, printing a full line of characters, time-after-time, advancing one character at the beginning of each line until the full character set is printed. The printer displays **TEST** while printing.
4. Press the **ON LINE** button again to stop printing. The printer again displays **WAIT**. If the **ON LINE** button is pressed again, the printer displays **LINE** and is online.

2.7.2 ASPI-46 Printer

After installing the printer, you can test its operation by using the self-test procedure. Before beginning the self-test, verify that

1. The paper and ribbon cartridge are inserted correctly.
2. The power cable is connected.
3. The front cover is closed.

Perform the following steps:

1. Press the **ON LINE** button while powering the printer on.

The display window says **TEST**, and a pattern is printed on the paper as shown in Figure 10-1 of Universal Station Service, in the Service binder. Note that the printer continues to print until you press the **ON LINE** button.

2. The display window says **WAIT**.

Service information including pinning, adjustments, tests, and troubleshooting are found in the *Universal Station Service* manual.

Printer operating and customizing instructions are also found in Appendix D of the *Process Operations Manual*, in the *Process Operations* binder.

2.8 REMOVABLE-MEDIA DRIVE CHECK

Your system contains high-capacity cartridge disk drives or floppy disk drives.

2.8.1 Cartridge Disk Drive Check

Prior to initial checkout, ensure that the ac power cable and the I/O cable are connected. Note that ac safety ground for the cartridge disk drive frame is made (connected) when the ac cable is plugged into the power entry box (except in USs that have the Acme power supply supplying power to the cartridge drives).

- If a polystyrene insert is in the cartridge slot, remove it carefully.
- Open the rear door of the console cabinet and apply power to the Universal Station.
- Ensure that the Cartridge Disk power supply DC Status LED is lit (Dual Node Module only).
- Perform System Validation, see Section 3.

Service information including pinning, adjustments, tests, and troubleshooting are found in the *Universal Station Service* manual. See Appendix A for handling and storage of the cartridge disk.

For instructions on changing the LCN module (node) address, refer to the *LCN System Installation Manual*.

2.8.2 Floppy Disk Drive Check

Prior to initial checkout, ensure that the ac power cable and the I/O cable are connected. Note that ac safety ground for the disk-drive chassis (frame) is made (connected) when the ac cable is plugged into the power-entry box (except in USs that have the Acme power supply supplying power to the cartridge drives).

- Before operating the 5 1/4 floppy disk drive, open the latch lever and remove the shipping insert. Do not attempt to close the latch lever without first inserting a diskette, or the shipping insert. To write protect the 5 1/4 diskette, the write tab must be taped over.
- Open the rear door of the console cabinet and apply power to the Universal Station.
- Ensure that the Disk Drive power supply DC STATUS LED is lit (Dual Node Module only).
- Perform System Validation, see Section 3.

For service information including pinning, adjustments, test, and troubleshooting, refer to the *Universal Station Service* manual in the *Service* binder. See Appendix A for handling and storage of the Flexible Diskette, commonly called the "floppy disk."

For instructions on changing the LCN module (node) address, refer to the *System Installation Manual*.

2.9 ANNUNCIATOR TERMINAL BOARD CONNECTIONS

All Universal Stations with an Operator keyboard include the external annunciator terminal board. This board is located in the console bay below the keyboard, on the lower-right panel.

The standard board, as shown in Figure 2-6, includes four solid-state relays, three of which can be used for switching external annunciators or indicators. The input connections to these relays are prewired so that actual determination of the relay that is activated by a specific alarm, event, or message priorities is made. This is achieved through configuration of SYSTEM WIDE VALUES.

For more information about Implementation, refer to the *Network Form Instructions* in the *Implementation/Startup & Reconfiguration -1* binder.

Ensure that terminal board to annunciator panel wiring matches the requirements (as provided in Figure 2-6).

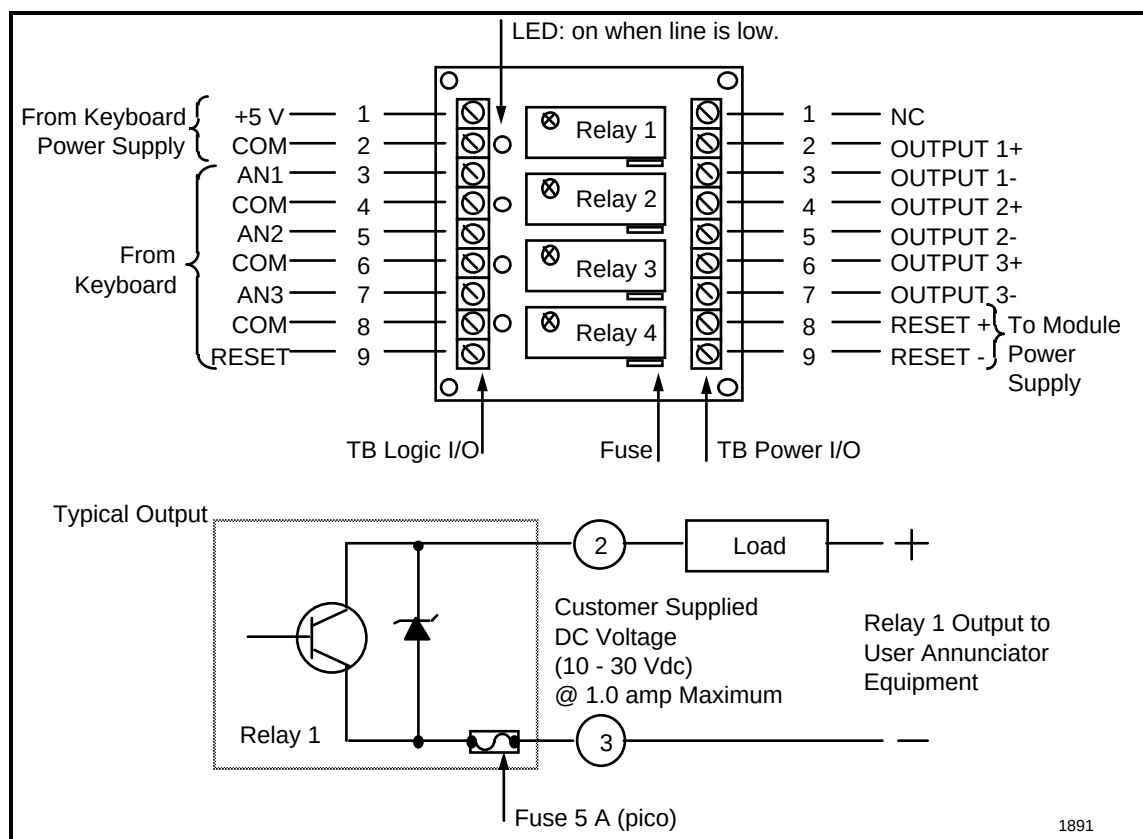


Figure 2-6 — External Annunciator Terminal Board

2.10 U^XS and A^XM Connection to the Network

The WSI I/O and WSI2 I/O boards connect to several different networks through different types of Media Access Unit (MAU). The MAU is mounted on the end of an adapter cable connecting to J6 on the WSI I/O board. It is important that the BNC or other connector type on the MAU does not come in contact with any conducting surface. This connection must be kept dry and must not be allowed to come in contact with contaminants.

3. SYSTEM VALIDATION

Section 3

This section lists all of the test programs and the initial configuration requirements.

3.1 HARDWARE VERIFICATION TEST SYSTEM (HVTS) OVERVIEW

The Hardware Verification Test System is first loaded on an empty system for post-installation system checkout. Later, it can be used as a troubleshooting aid on nodes physically separated from the on-process LCN. Test-system loading and set-up instructions are provided in the HVTS publications listed in subsection 1.2.

For instructions on Diskette/cartridge handling and loading, see Appendix C of the *Process Operations Manual*, in the *Process Operations* binder.

A successful pass through each of the following subsystem test programs ensures that the system hardware and firmware elements are functional, and that the system is ready for the operational software required for system startup:

- Local Control Network Exerciser (LCNX)
- Keyboard Subsystem Tests (KEYB)
- Operator Interface Subsystem Tests (OPIF)
- Smart Controller Magnetic Disk (SCMD)
- Winchester Disk Subsystem Tests (WINC)
- Floppy Disk Subsystem Tests (FLPY)
- Data Hiway Interface Subsystem Tests (DHIF)
- System Input/Output Subsystem Tests (SIOS)
- Memory Exerciser Module Tests (MEMX)
- Communications Line Interface Tests (CLIF)
- Data Hiway Exerciser Tests (DHEX)
- System Clock Exerciser Tests (CLKS)
- PIC/VDG Subsystem (PDGI)
- Smart Controller Magnetic Disk Tests (SCMD)
- Applications Module Redundancy Tests (AMRT)
- Process Network Interface Tests (PNIF)

- Programmable Logic Controller Interface Test (PLCT)
- Network Gateway Interface Test (NGIF)
- Process Network Interface Enhanced Test (PNIE)
- LCN Sequence of Events (LSOE)
- General Gateway Serial Interface Tests (GGSF)
- LCN US with X-Windows Workstation Tests (USXF)
- Workstation Interface Test (WSIF)
- Application Module^X Interface Tests (XAMF)

Note that the section numbers refer to corresponding sections of the HVTS publications listed in subsection 1.2.

3.1.1 CMTS Tests for GUS

When power is applied to a GUS, it runs its self-tests. After successful completion of the self-tests, GUS stations require successful completion of the Core Module Test System PCIF Test and completion of the self-tests that run when power is applied.

3.1.2 Configuration and Revision Information

After the HVTS program is loaded, we recommend you to enter the command "CON" for "configuration" to obtain a printout of the node under test. This printout provides you with a record copy of the node hardware configuration.

An example is shown in Figure 3-1. In the example, entries made by the operator are enclosed in boxes. In addition, the operator steps through three different nodes obtaining the configuration for each.

The following figure illustrates a Configuration Sample Printout:

```

○ 11:05 NODE 1 ALL ?CON ○
19-SEP-1989 11:05:22 Node 1 Universal Station
○ Slot 1 HPK2 4004, 000A and 300000 Memory Words At 080000 ○
Slot 2 LCNI 4758, 05A2, 4758, 0593, 0200, 8806, 8000, 0101
○ Slot 3 SPDG 44CA, 1F02, 0098, 0099, 0200, 8400, 000E, 0000 ○
Slot 4 EPDG 4442, , 640D, F0A3, 0202, , 0F00, 0101
○
○ 11:05 NODE 1 ALL ?NODE 5 ○
○ 11:05 NODE 5 ALL ?CON ○
19-SEP-1989 11:05:30 Node 5 Application Module
○ Slot 1 HPK2 4003, 0008 and 200000 Memory Words At 080000 ○
Slot 2 LCNI 4C58, 05A2, 4C58, 0591, 0200, 8C06, 010E, 0101
○
○ 11:05 NODE 5 ALL ?NODE 6 ○
○ 11:05 NODE 6 ALL ?CON ○
19-SEP-1989 11:05:39 Node 6 History Module
○ Slot 1 HPK2 4203, 000A and 200000 Memory Words At 080000 ○
Slot 2 LCNI 4C58, 05A2, 4C58, 0591, 0200, 8C06, 010E, 0101
○ Slot 3 SPC 43C1, 0700, 0798, 0798, 0500, 4000, 2003, 0101
○
3018

```

Figure 3-1 — Configuration Sample Printout

NOTE

You may notice some apparent discrepancies between the printout obtained in the preceding section and the physical boards actually found in the equipment. For example, software cannot tell the difference between an LCNI and an LLCN board. Although your equipment may use an LLCN, the software will report an "LCNI." In addition, software is made to "think" a physical EPDG board occupies two board-slots. Software reports the first "slot" is occupied by an "SPDG" board and the second nonexistent "slot" contains an "EPDG."

Incidentally, the first slot above an EPDG **cannot be physically occupied by any board** and the second slot above an EPDG can be occupied by **only certain boards**. See the *Five/Ten-Slot Module Service* manual in the *Service* binder for more detailed information.

If you enter the "REV" command, the revision of the hardware (HW) and firmware (FW) is displayed as illustrated in Figure 3-2. Again, entries made by the operator are enclosed in boxes in this example. The operator again steps through three nodes, obtaining the revisions for each.

A. OPERATING PRACTICES

Appendix A

A.1 OPERATING PRACTICES AND HOUSEKEEPING

Listed here are some do's and don'ts pertaining to operating practices and general housekeeping that must be followed during startup and normal everyday operations.

A.1.1 Before Startup

1. Thoroughly clean all operating areas, subfloor areas, cable raceways, heating and air-conditioning ducts, and plenums.
2. Make sure that all control-room windows are sealed.
3. Place impregnated mats at each entrance to a control area to prevent dirt and dust from being tracked in.
4. Provide a coat rack and/or closet outside the operating area for removal of any outer clothing made of nylon or other synthetic fabrics, except where flame-retardant uniforms are mandatory at all times.
5. Make sure that the furniture and carpets are not made of materials that can combine with clothing to create static electricity.
6. Prepare a regular cleaning schedule for specific area requirements and for cleaning of consoles, cabinets, and peripheral devices where necessary.

CAUTION

Do not attempt to clean the printed-wiring boards.

7. Establish a "no smoking" rule in the operating area. Smoke and other fine dust particles can damage cartridge and floppy disks and their drives.
8. When swapping or handling printed-circuit cards, use a static-control device, for example, a wrist strap, see CIRCUIT CARD HANDLING, Appendix A.

A.1.2 After Startup

1. Continue your "no smoking" rule in operating areas that contain magnetic tape, floppy, cartridge or hard disk drives, or where disks and media are stored.
2. Maintain humidity levels (ideally) between 40 and 60%. (Lower humidity may cause static-discharge problems.)
3. Control humidity fluctuations to a rate-of-change less than 6% per hour.

4. Do not defeat temperature and humidity controls by opening doors and windows (for example, to enhance operator comfort).
5. Keep traffic in the control-room operating areas to a minimum. Restrict access to authorized personnel, whose duties require control room entry.
6. Review procedures for extinguishing electrical fires and establish fire-fighting procedures. Refer to a qualified fire fighting systems contractor for assistance.
7. Plant personnel frequently use hand-held radios ("walkie-talkies"), or citizen-band radios mounted in maintenance vehicles for communication. To avoid RFI problems, review procedures for proper use of such devices as follows:

If radio communication must take place within an operating area or input/output area, a base-station transceiver with an external antenna must be used.

For other applications, radio transmitters with outputs rated as high as 5 Watts must be kept at least 3 meters (10 ft) from the TPS equipment during operation. Transmitters with outputs higher than 5 Watts must be kept as far as possible from your consoles and cabinets. (Refer to *Data Hiway Subsystem Site Planning* for distance formula.) Keep equipment doors closed.

Other sources of RFI include generators, arcing relays, or motor contacts, and so on.

8. Follow proper cleaning procedures when cleaning the operator area or the control room:
 - Do not use water freely. Mop must be only dampened, not wet or dry.
 - Use a lint-free, antistatic-type dust cloth to remove dust.
 - Do not sweep around areas containing magnetic disk or diskette drives.
 - Use vacuum cleaner on carpets—preferably one connected to an external system.
 - Do not allow liquids to be placed on the Universal Station keyboards. Liquid spills damage electronic equipment.
9. Clean the floppy disk/cartridge disk drive (if required) as outlined in the *Universal Station Service* manual in the *Service* binder, to prevent error display when loading programs.
10. Regularly clean the CRT face and touchscreen to minimize operator fatigue. Cleaning procedures are found in the *Universal Station Service* manual in the *Service* binder.
11. Clean or replace cabinet/module ventilating-fan filters once a month in dirty environment, every two or three months in clean environment (for example, a control room or operating center that does not open into a dirty area is considered a clean environment). Use only Honeywell-recommended replacement filter or equivalent. Follow filter-cleaning instructions provided in the *Universal Station Service* manual in the *Service* binder.
12. Clean the printer and trend pen recorder before startup and periodically thereafter, as described in the *Universal Station Service* manual in the *Service* binder.
13. Periodically clean the operator and engineer keyboards by dampening a cloth with mild detergent and wiping the keys. Do not spray detergent solution on keys as leakage may ruin the circuits underneath.

A.2 FLEXIBLE DISKETTE HANDLING AND STORAGE

Upon receipt of Flexible (floppy) Diskettes, inspect them for signs of shipping damage. Allow at least one hour for temperature adjustment to the computer-room environment before using them.

Recommended storage is on shelves in the computer or control room. They must be kept in their protective storage envelopes at all times except when installed in the diskette drive. Store the diskettes flat (horizontal) in stacks of no more than ten.

CAUTION

Never touch or attempt to clean the areas exposed by the openings in the flexible-diskette jacket. Ensure that the flexible diskette is kept in its protective envelope while you write on the label.

The system diskettes are provided with labels attached to the jacket. Writing on the labels is permitted but only with a felt-tipped pen. Do not use ball point pens, lead pencils, grease pencils, or any marker that could dent the diskette jacket or leave loose contamination.

A.2.1 Handling and Loading

- When loading a diskette, ensure that the FD drive is powered on.
- After opening the drive's door or latch lever, remove the diskette from its protective envelope. Then carefully slide the diskette into the drive until the jacket rests against the stops and close the drive's door or latch lever.
- Always return the empty protective envelope to the diskette storage-box or other clean area and protect it from liquids, dust, and other contaminants.
- Replace the diskette in its protective envelope and return it to a storage box whenever it is not in use.
- Never leave a diskette partially inserted in the drive.
- Never use paper clips on a flexible diskette cartridge.
- Inspect each diskette during normal operation and remove damaged or contaminated diskettes from service.

A.3 CIRCUIT CARD HANDLING

The circuit cards or Printed Wire Boards (PWBs) are adequately protected from damage caused by Electrostatic Discharge (ESD). This is possible only when they are installed in the system module or card file, or packed inside the conductive plastic bag in which they are shipped. To avoid ESD damage when the card is handled outside its enclosure, to guard against electrical overstress, and to maintain personnel safety, the following practices and procedures must be followed:

- Turn off power to the module or card file before removing or inserting the card.
- Handle the card only by its edges. Do not touch the lands, connectors, or components unless you are wearing a grounded wrist-strap and the card is on a conductive work-surface.
- When applying power to a system before installation is complete, terminate all loose wires within the cabinet or console. Make sure power is turned off when any wiring is being done.
- ESD-generating materials, such as plastic, rubber, nylon, polyester, vinyl, silk, or synthetic materials or garments, must not be allowed in the area of the cards. If you are wearing clothing of such material, you must stand on a grounded floor-mat while wearing a grounded shoe-strap, or you must wear a grounded wrist-strap while handling cards. Note: Take special care to always keep the cards away from the material because static charges cannot be drained off, except by discharge.
- Do not carry unprotected cards across carpeting, unless it is grounded conductive carpet such as conventional fiber with woven-in ground wire. Always keep the circuit card in its protective bag until it is actually needed.
- All test equipment and tools must be connected to the metal card-file chassis or module frame with a ground wire, before touching the card or internal wiring.
- Cards must be handled and transported to and from the job site in their protective bags (see approved material list).
- Personnel must wear an approved wrist-strap connected to the chassis before removing the card from its protective bag or card slot.
- When shipping a suspected defective card, pack it in its protective bag before placing it in the shipping carton. Note that cards must be protected against further damage so that failure analysis can be accomplished.
- Do not use standard bubble-pack mailers.
- Do not allow unprotected cards to come in contact with styrofoam packing material.

For additional ESD information, refer to the *LCN Site Planning Manual* in the *LCN Site Planning & Installation* binder.

A.4 CARTRIDGE DISK HANDLING AND STORAGE

Upon receipt of Cartridge Disks, inspect them for signs of shipping damage. Allow at least one hour for temperature adjustment to the computer-room environment before using them.

Recommended storage is on shelves in the computer or control room.

Although the cartridge protects the disk from most accidental damage, the following cartridge handling rules must be observed:

- Do not try to open the cartridge when it is outside the drive.
- Do not insert objects into the cartridge.
- Remove the cartridge from the drive after use and store in its protective jacket.
- Do not expose the cartridge to direct sunlight or moisture.
- Do not expose the cartridge to magnetic fields greater than 30 oersteds.
- Protect the cartridge from dirt, spills, and harsh environment.
- Avoid handling the front edge of the cartridge, because oils can be transferred from the hands to the cartridge disk.

A.4.1 Loading

- Drive power must be turned on before the cartridge can be inserted. If power to the drive is turned off, the latch pin in the drive prevents cartridge insertion.
- When the cartridge is inserted into the drive, shutter first. The cartridge edge containing the shutter faces toward the stop button on the front of the drive (see Figure A-1). Interlocks in the system prevent improper cartridge insertion.

CAUTION

If the cartridge is difficult to insert, check its orientation and try again. Do not force the cartridge into the drive. Objects inserted into the cartridge opening in the front of the drive may cause damage to the drive. Such an action voids the manufacturer's warranty.

- After loading a cartridge, the green indicator on the front of the drive, next to the stop button blinks as the motor picks up speed. When the indicator stops blinking and stays lit, the drive is READY. See Figure A-1.

NOTE

The yellow LED flashes briefly as the drive reads initialization information from the disk. If the drive fails to initialize correctly, an error condition exists. If this occurs, reinsert the cartridge to ensure proper seating. If this does not correct the problem, the media may require reformatting or the drive may require service. See Figure A-1.

- To remove the cartridge, push the stop button. The green LED then begins to blink as the motor spins down. When the motor comes to a stop, the green LED turns off and the latch pin disengages. The cartridge can then be removed. See Figure A-1.

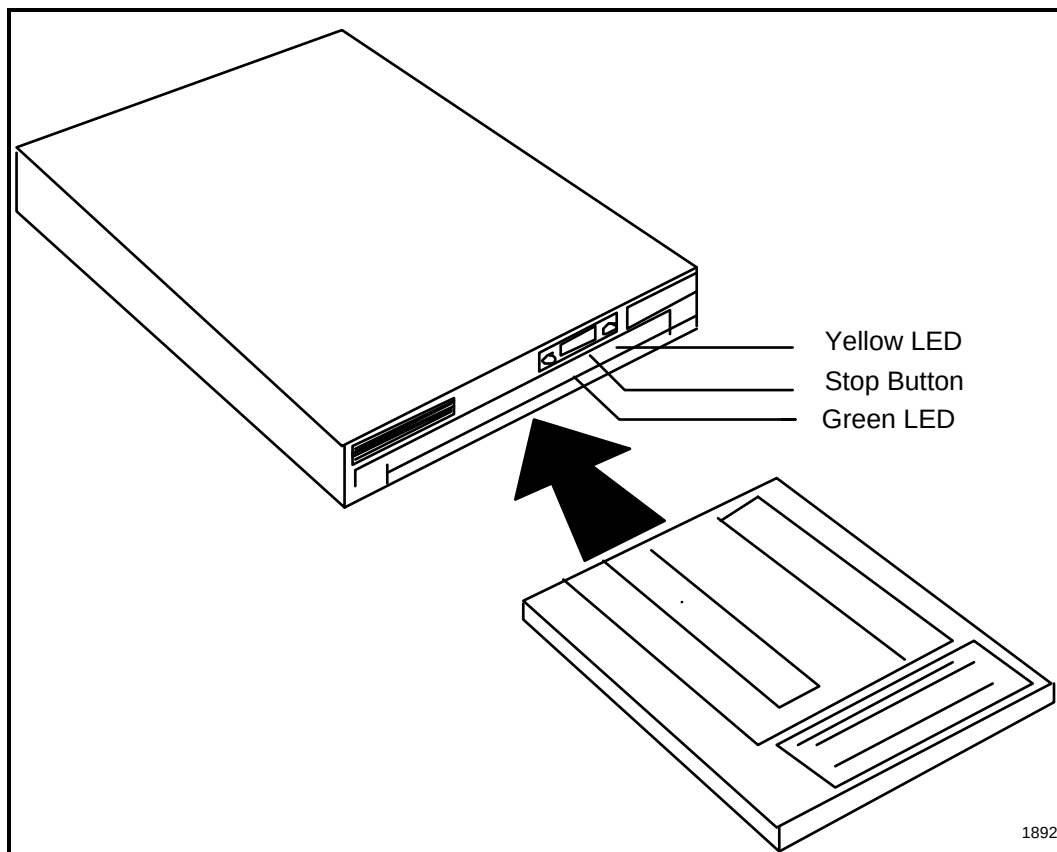


Figure A-1 — Cartridge Insertion

B. CLIPS AND SWITCHES (SYSTEM PINNING)

Appendix B

All LCN modules, hiway boxes, and peripheral devices require a unique system device address.

For system module characterization (node-address pinning) and Hiway Gateway preferred access cabling, refer to *LCN System Installation* manual, in the *LCN Installation* binder. For peripheral-equipment characterization (option selection), refer to the *Universal Station Service* manual in the *Service* binder. For Winchester Disk drive characterization and placement of the terminator DIP, refer to the *History Module Service* manual, also in the *Service* binder.

C. TPS LCN CABLE TESTING (USING TEKTRONIX TDR 1502/1503)

Appendix C

C.1 PREPARATION

Read this procedure and have all the test equipment and material available. See subsection 1.3 TOOLS AND TEST EQUIPMENT.

Cable sections of less than 3 feet must be tested with the Model 1502 tester. The 1503 tester has a minimum resolution of 3 feet.

C.2 SETUP

Set up the TDR Cable Tester as follows:

POWER		:	ON or OUT	
RETURN LOSS (0 dB to 60)		:	0 dB	(or SENSITIVITY)
RETURN LOSS (FINE)		:	0 dB	(or SENSITIVITY)
FEET/DIV (RED) (X10/X100)		:	X10	
FEET/DIV (50 to .5)		:	.5	
NOISE FILTER		:	OUT	
POSITION & FINE		:	BASE LINE IN CENTER	
INTENSITY		:	AS REQUIRED	
FOCUS		:	AS REQUIRED	
DISTANCE		:	000	
IMPEDANCE	50 Ohms	:	OUT	
IMPEDANCE	75 Ohms	:	IN	
IMPULSE WIDTH	10 ns	:	IN	(or SHORT PULSE)
IMPULSE WIDTH	100 ns	:	OUT	(or MEDIUM PULSE)
IMPULSE WIDTH	1000 ns	:	OUT	(or LONG PULSE)
CABLE DIELECTRIC	SOLID POLY	:	OUT	

CABLE DIELECTRIC	FOAM POLY	:	IN	(or .79 DISTANCE CAL)
CABLE DIELECTRIC VAR		:	AS IS	
ZERO REF (SET)		:	Put First Pulse on Left of Screen	
0 dB SET		:	AS IS	

C.3 TEST PROCEDURE

NOTE

This test procedure does NOT work with multiple pieces of COAX cable connected together.

Test each length of cable, one at a time, as follows:

1. With no COAX connected, the TDR screen must look like Figure C-1.
2. Connect the COAX to be tested to the Cable Jack on the TDR. Do not terminate the other end. The screen must look like Figure C-2 or C-3.
3. Using both the ZERO REF and DISTANCE controls, position the second pulse with its peak on the vertical center line on the screen. Use POSITION to center the horizontal base line.
4. The cable terminator can be put on or taken off several times to verify that it is the "returned" pulse and not the "transmitted" pulse that is in the center of the screen.
5. With the terminator off and using the "RETURN LOSS fine" (and if necessary, "0 dB SET"), adjust the return pulse to three divisions high while keeping the base line on the horizontal center line of the screen.
6. Ensure that the screen looks like Figure C-4 or Figure C-7.
7. Connect the "Selected Terminator" (not just any terminator), using the "jack-to-jack" adapter or barrel splice, to the free end of the COAX.
8. Ensure that the screen looks like Figure C-5 or Figure C-8.
9. Take note of the RETURN LOSS (dB) settings, add 34 dB of gain to the setting and then adjust the RETURN LOSS (dB) knobs to the new calculated setting.

 Example 1: 0 on main, 0 on fine; read just to 30 on main, 4 on fine.

 Example 2: 0 on main, 5 on fine; read just to 30 on main, 9 on fine.
10. Adjust the POSITION and FINE controls until the left and right base lines are on the center line of the screen as in Figure C-6 or Figure C-9.

11. If any part of the return pulse (the same width as the unterminated pulse) is three or more divisions from the base line (in either direction), the COAX is REJECTED, as in Figure C-10. If the return pulse is less than + or - three divisions, the COAX is ACCEPTED, as in Figure C-11.

If any part of the trace between the transmitted pulse and the returned pulse is three or more divisions (+ or -) from the base line, REJECT the COAX under test.

Both ends of every coax must be tested. If one end measures three or more divisions from the base line, the COAX is REJECTED.

12. Tag all REJECTED COAX and perform necessary steps to replace them.

NOTE

If Steps 1 through 10 are done very carefully, Step 11 can be repeated many times (for the same length of COAX) without any adjustments, and changing only the "COAX under test," for assembly line type testing.

C.4 TEST COMMENTS/CONDITIONS

- A. Keep batteries on the 1503 TDR Cable Tester fully charged.
- B. Do not compromise on the accuracy of the terminator. This whole procedure to ACCEPT or REJECT the coax depends on the delta or difference between the SELECTED TERMINATOR and the COAX UNDER TEST. Paint or identify the SELECTED terminator because it is hard to find in a normal distribution of terminators.

Terminator specifications allow a 2% variation, that is, ± 1.5 ohms or 73.5 to 76.5 ohms. The selected terminator must be within 0.1 ohm, that is, 74.9 to 75.1 ohms. Several years ago, some terminators were mistakenly made with inductive resistors that measure properly on an ohmmeter, but show as "s" rotated 90 degrees on the TDR, see Figure C-12. Do not use such terminators on the LCN or Data Hiway.

- C. If absolutely necessary, you can use a "T-connector" instead of the "Splice" to connect the COAX to the terminator. However, the "T-connector" must not have the 1-ohm resistor in series or the 56 k ohm resistor to the case of the "T-connector."
- D. Workmanship: The BNC connectors must be installed on the COAX with the correct tools. If the connection was made using PLIERS, CHANNEL-LOCKS, or VICE GRIPS, REJECT the COAX. Check for folds or kinks in the COAX, or whether the minimum bend radius is exceeded.
- E. When the COAX and TERMINATOR match each other, the returned pulse is very small or very flat on the base line. Figures C-6, or C-9, or C-11 look almost flat at the returned pulse.
- F. When the two "dB" values are mentioned, the first refers to the main setting; that is, 0, 10, 20, 30, 40, 50, or 60 dB. The second value refers to the variable or fine setting; that is, 0 to 18 dB. Example: 30 dB and 8 dB means 30 on the main setting and 8 on the variable setting for a total gain of 38 dB.
- G. The *BASIC System Checkout Manual*, (4/84 or later), has a "Data Hiway" TDR Cable Acceptance Test. Using + or -3 ohms (or 2.03%), this test can be used for checking the middle of LCN cables.

The following figures illustrate the returned pulse:

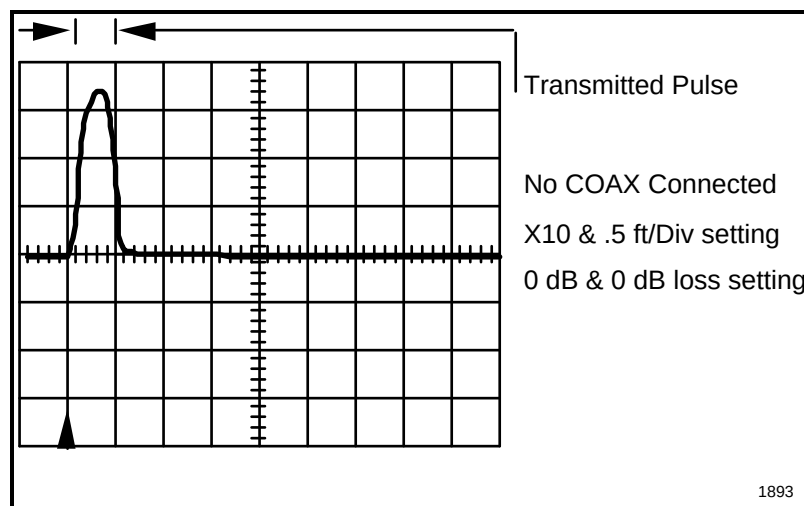


Figure C-1

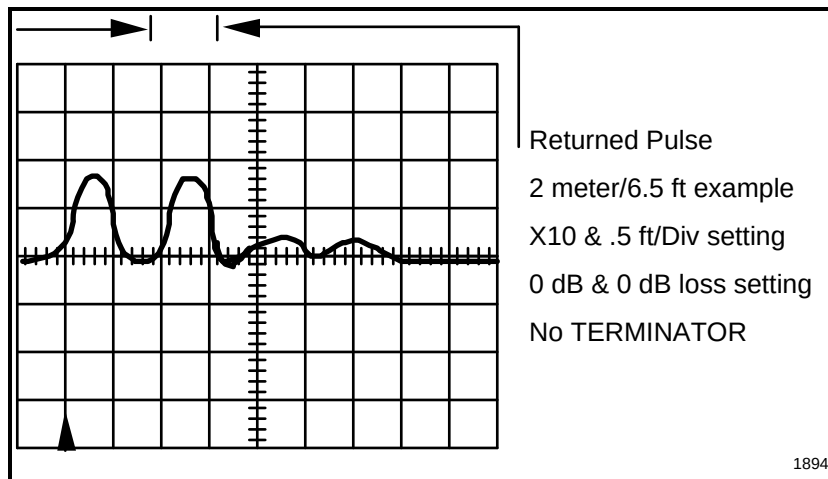


Figure C-2

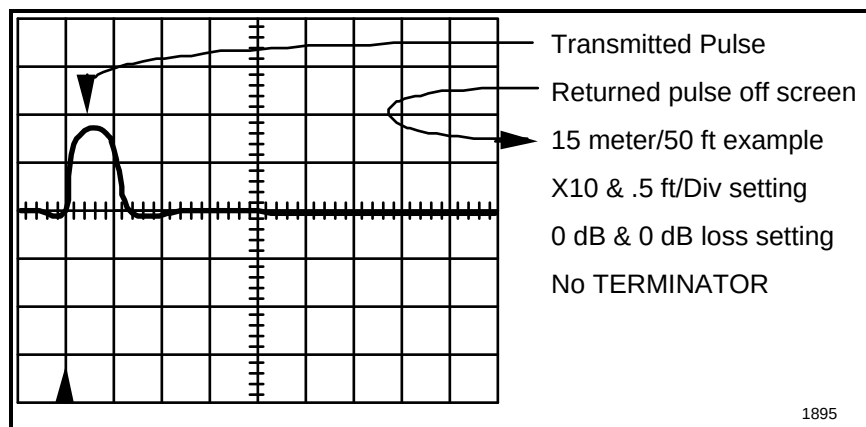


Figure C-3

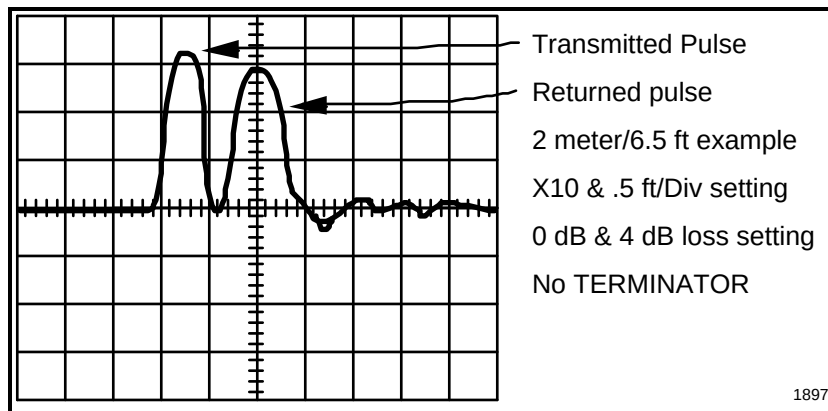


Figure C-4

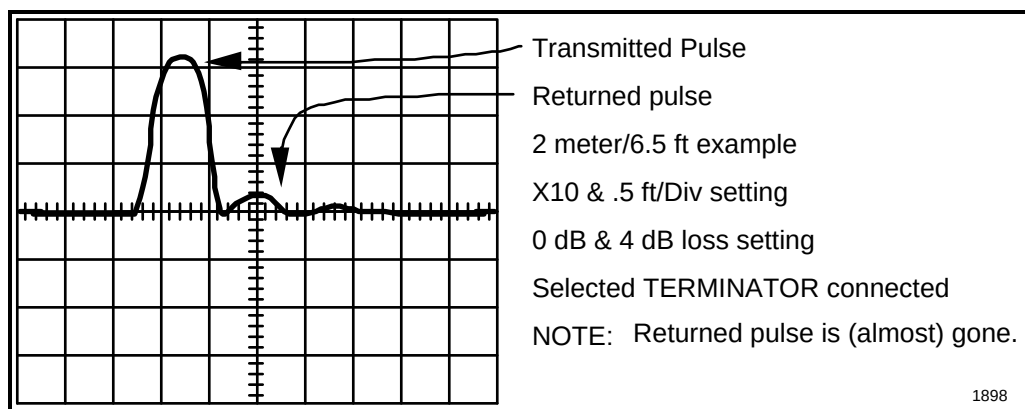


Figure C-5

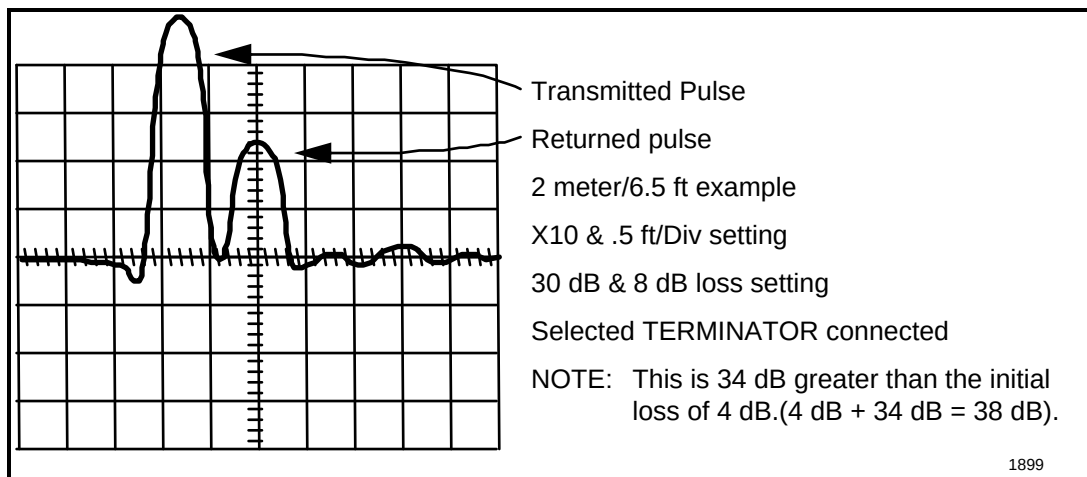


Figure C-6

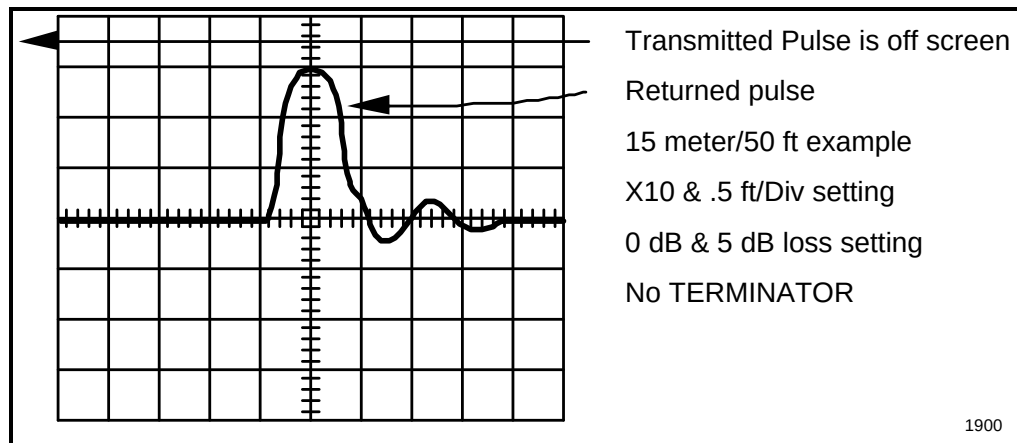


Figure C-7

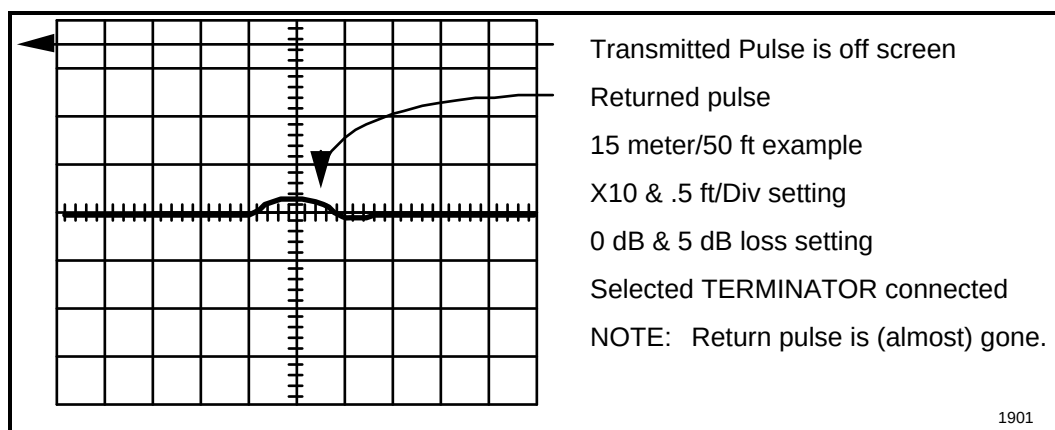


Figure C-8

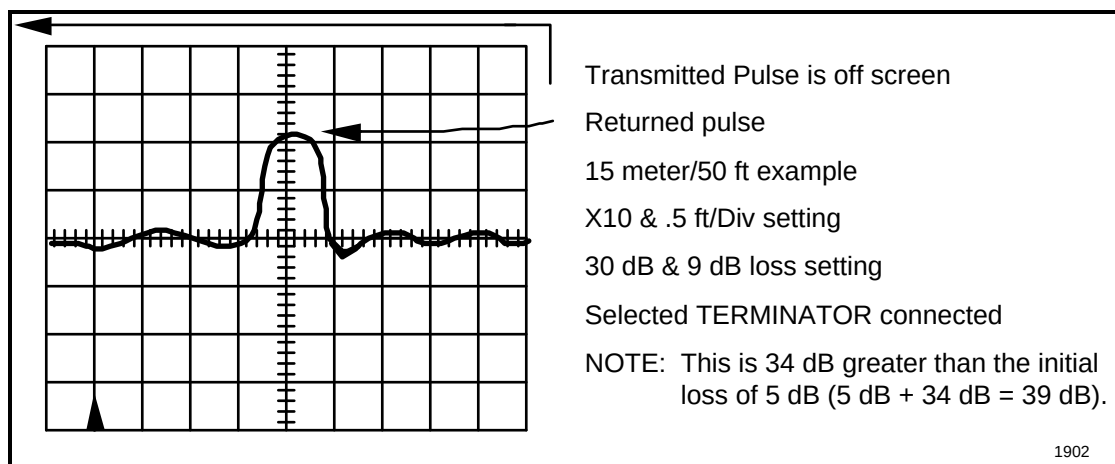


Figure C-9

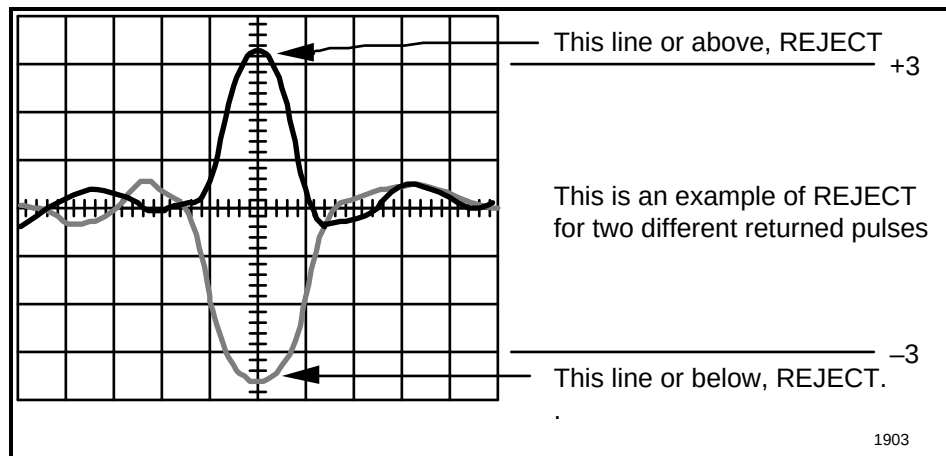


Figure C-10

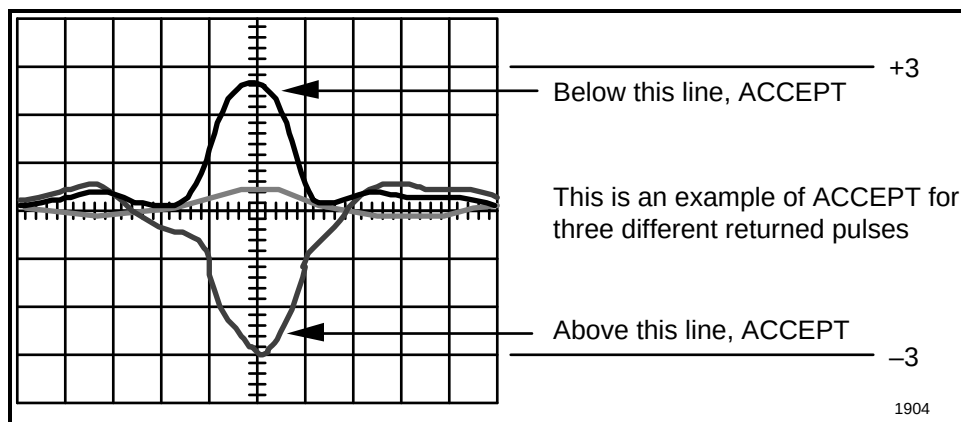


Figure C-11

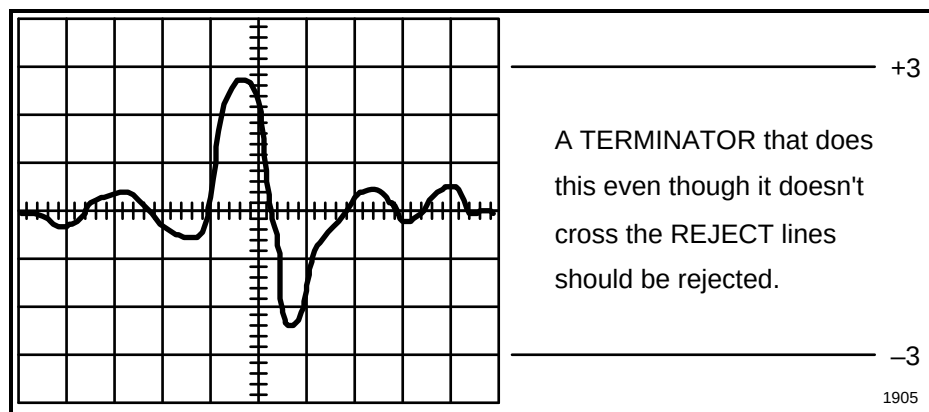


Figure C-12

C.5 EQUATIONS AND SAMPLE CALCULATIONS

EQUATIONS

Reflection Coefficient = Greek Letter "RHO," shown here as the letter "p"

$$1. \quad p = \frac{E \text{ return}}{E \text{ transmitted}}$$

$$2. \quad p = \frac{(Z \text{ test} - Z \text{ ref})}{(Z \text{ test} + Z \text{ ref})}$$

$$3. \quad Z \text{ test} = Z \text{ ref} \frac{(1 + p)}{(1 - p)}$$

$$4. \quad |p| = 10^{-\frac{(\text{RETURN LOSS dB})}{(20)}}$$

$$5. \quad \text{RETURN LOSS}_{\text{dB}} = 20 \text{ LOG}|p| \quad |p| \text{ means absolute value of } (p)$$

SAMPLE CALCULATIONS WHEN THE TERMINATOR VARIES

	(COAX)	(TERMINATOR)	(PULSE)			RETURN
	Z ref	Z test	polarity	p	%	LOSS dB
NG	75	INF	+	1.0000	100.000	0.0000
NG	75	1.0 Meg	+	0.9998	99.985	-0.0013
NG	75	1.0 k	+	0.8604	86.046	-1.3053
NG	75	225.0	+	0.5000	50.000	-6.0205
NG	75	100.0	+	0.1428	14.285	-16.9019
OKAY	75	78.12	+	0.0203	2.037	-33.8175
OKAY	75	75.10	+	0.0006	0.066	-63.5276
OKAY	75	75.00	None	0.0000	0.000	-INF
OKAY	75	74.90	-	0.0006	0.066	-63.5160
OKAY	75	72.00	-	0.0204	2.040	-33.8036
NG	75	50.0	-	0.2000	20.000	-13.9794
NG	75	25.0	-	0.5000	50.000	-6.0205
NG	75	0.0	-	1.0000	100.000	0.0000

SAMPLE CALCULATIONS WHEN THE COAX VARIES

	(COAX)	(TERMINATOR)	(PULSE)			RETURN
	Z ref	Z test	polarity	p	%	LOSS dB
OKAY	78.12	75	-	0.0203	2.037	-33.8175
OKAY	75.10	75	-	0.0006	0.066	-63.5276
OKAY	75.00	75	None	0.0000	0.000	-INF
OKAY	74.90	75	+	0.0006	0.066	-63.5160
OKAY	72.00	75	+	0.0204	2.040	-33.8039

Notice: In both sets of SAMPLE CALCULATIONS, the pulse is positive if the TERMINATOR is GREATER than the COAX and the pulse is negative if the TERMINATOR is LESS than the COAX.

D TPS SYSTEM POST-INSTALLATION CHECKLIST

APPENDIX D

Customer_____SID_____Date(delivery)_____Date(Start up)_____

POWER OFF CHECKS*

A. Grounding Checks (power off).

AC Safety Ground.

Master Reference Ground (MRG).

Reference *TPS System Site Planning Manual*, *SY-09-03 (Basic System Check-Out Manual)*, *SY-20-04 (DH Site Planning Manual)*, and *Process Manager Site Planning*.

B. Hiway/LCN Cable Separation Rules.

Ensure that cable A is cable "A" to all boxes/nodes, also that cable B is "B".

To all boxes/nodes, see item G, next sheet.

Reference *SY-20-04 (DH Site Planning Manual)* and *HO-20-01 (DH Cabling, Wiring and Interconnection)*.

C. Conduct Time Domain Reflectometer tests on the Data Hiway (DH) and Local.

Control Network (LCN) cables, as required. See Appendix C for LCN cables.

Reference *SY-09-03 (Basic System Check-Out Manual)* for the Data Hiway cables, and *Process Manager Checkout* for UCN cables.

D. Hiway/LCN TEEs and Terminators.

Check for proper installation and type. TEEs are not interchangeable.

Reference *SY-20-04 (DH Site Planning Manual)*, *PI-09-01 (PIU Installation Manual)*, *Process Manager Installation*, and *LCN System Installation Manual*.

E. Preferred Access Cable wiring.

Check cabling between preferred-access device and HTD for proper shield.

Ground and addressing priority.

Reference *SY-20-04 (DH Site Planning Manual)* and *PI-09-01 (PIU Installation Manual)*.

F. Check isolation of intrinsically safe wiring.

Reference *SY-20-04 (DH Site Planning Manual)* and *PI-09-01 (PIU Installation Manual)*.

Customer_____SID_____Date(delivery)_____Date(Start up)_____

POWER ON CHECKS*

A. Verify operation of all cabinet/module fans.

Reference SY-09-03 (*Basic System Check-Out Manual*), *PM Service Manual*, and *Five/Ten-Slot Module Service*.

B. Verify proper dc-voltage levels on all Basic/PIU power supplies, including

bulk supplies and power-regulator cards.

Reference applicable service manual(s).

*See subsection 1.2 for the list of support/reference documents.

C. Check ac-line levels and stability, using an ac-line analyzer to record incoming

Voltage spikes, sags, frequency disturbances, and durations of same.

Reference SY-09-03 (*Basic System Check-Out Manual*) .

D. Check file/module for proper power-up LED display.

Reference SY-09-03 (*Basic System Check-Out Manual*) , *PM Service Manual*, and *Five/Ten-Slot Module Service* in the *Service binder*.

E. Verify hiway-device addressing and polling response.

Reference SY-09-03 (*Basic System Check-Out Manual*), *HLPIU-M (HL PIU Maintenance Manual)*, and *Hardware Verification Test System*.

SYSTEM OPERATIONAL CHECKS

Customer_____SID_____Date(delivery)_____Date(Start up)_____

A. OP Station Diagnostic Checkout (if present).

Shake and tap Hiway TEEs with test running.

Reference SY-09-03 (*Basic System Check-Out Manual*).

B. Check operation of all optional BASIC equipment and peripherals (if present).

Reference SY-09-03 (*Basic System Check-Out Manual*).

C. Run Universal Station serial-printer unit diagnostic.

Reference *Universal Station Service* manual in the *Service binder*.

D. Check operation of Trend Pen Recorder.

Reference RO-02-00 (*Recorder Service Data*) in *Universal Station Service Addendum*.

E. CMTS (P/N 51150354).

Reference *Core Module Test System*.

F. LCNI Network Communications Test (P/N 51150328).

Reference *LCNI Network Communications Test*. This test can be used to verify that LCN cables are not crossed at any nodes.

G. Hardware Verification Test System (HVTs).

Refer to subsection 1.2 (P/N 51105670).

- _____ LCNX LCN Exerciser
- _____ DHEX Data Hiway Exerciser
- _____ DHIF Data Hiway Interface Subsystem tests*
- _____ FLPY Floppy Disk Subsystem tests
- _____ KEYB Keyboard Subsystem tests
- _____ MEMX Memory Exerciser Module tests
- _____ OPIF Operator Interface Subsystem tests
- _____ SIOS Status I/O Subsystem tests
- _____ CLKS Clock Subsystem tests
- _____ CLIF Communications Line Interface tests
- _____ PDGI Pick/VDG Subsystem tests
- _____ SCMD Smart Controller Magnetic Disk tests (limit test to read only)
- _____ AMRT Application Module Redundancy tests
- _____ PNIF Process Network Interface tests
- _____ PLCG Programmable Logic Controller Interface tests
- _____ CNIF Communications Network Interface tests
- _____ NGIF Network Gateway Interface tests
- _____ PNIE Process Network Enhanced tests
- _____ LSOE LCN Sequence of Events test
- _____ GGSF Generic Gateway Serial Interface test

_____ USXF LCN US with X-Windows Workstation tests (U^XS only)

_____ WSIF Workstation Interface test (U^XS only)

H. Verify LCN Node and Data Hiway Box addresses (see Appendix B).

Develop Hiway/LCN box address/location map.

- I.** If a communication network is connected to one of the following node types, the contractor who designed and installed the network must check out the network.

*Do not run the DHIF test on an ON-LINE Hiway or with the HLPIU or CB connected to the process.

E. TPS LCN CABLE TESTING (USING RISER BOND TDR 1270A)

Appendix E

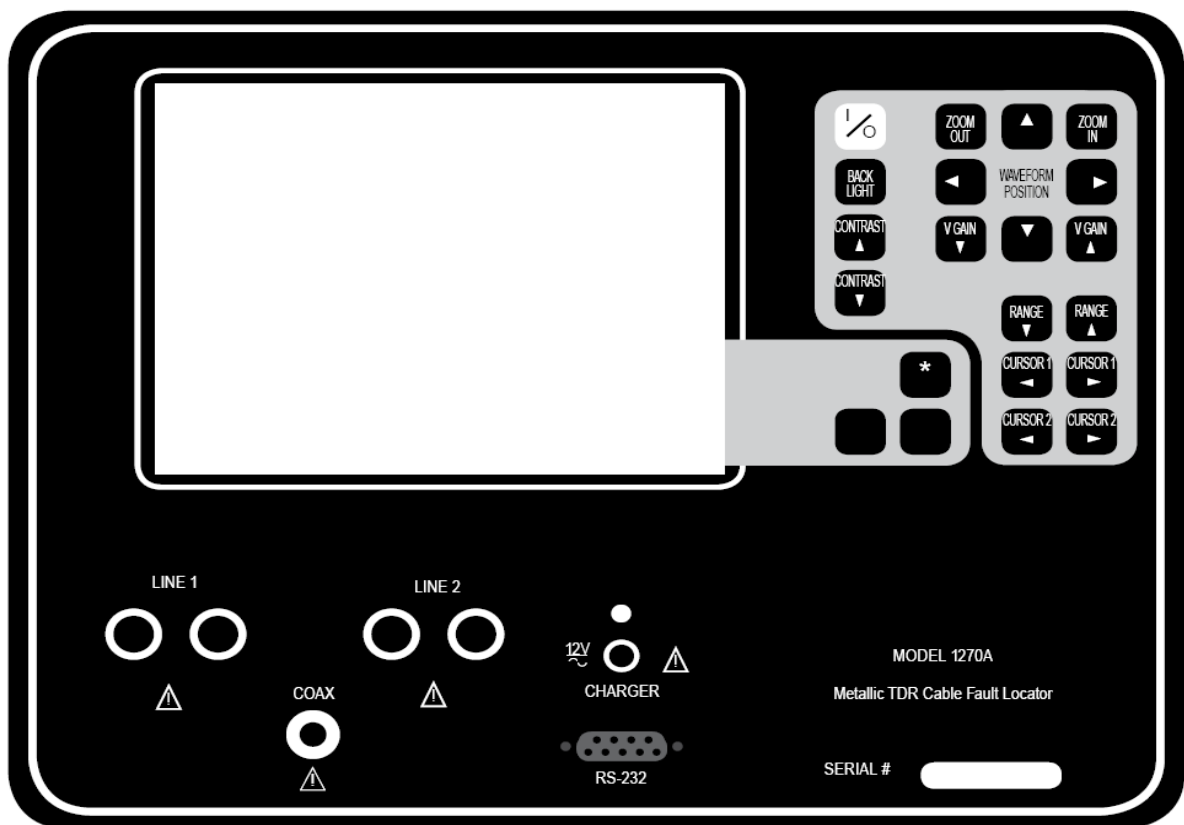
E.1 PREPARATION

Read this procedure and have the test equipment and material available. See subsection 1.3. TOOLS AND TEST EQUIPMENT.

Cable sections of less than 10 meters must be tested with a pulse width of 2 ns. Cable sections of larger than 10 meters must be tested with a pulse width of 25 ns.

E.2 SETUP

E.2.1 Front Panel Description



E.2.1.1 Keypad**On/Off:**

I/O Use the I/O key to turn the instrument on and off.

Backlight:

Use the backlight key to turn the CFL backlight on or off.

Contrast:

Use the two arrow keys to change the contrast of the LCD.

Zoom In, Zoom Out:

Use the two zoom keys to zoom in or out on an area of interest on the waveform display.

Waveform Position:

Use the four arrow keys to move the position of the waveform(s) left, right, up, and down.

V Gain:

Use the two arrow keys to decrease and increase the vertical waveform amplitude or gain.

Range:

Use the two range keys to increase and decrease the cable distance displayed on the screen. Pulse width and vertical gain are automatically adjusted for each range.

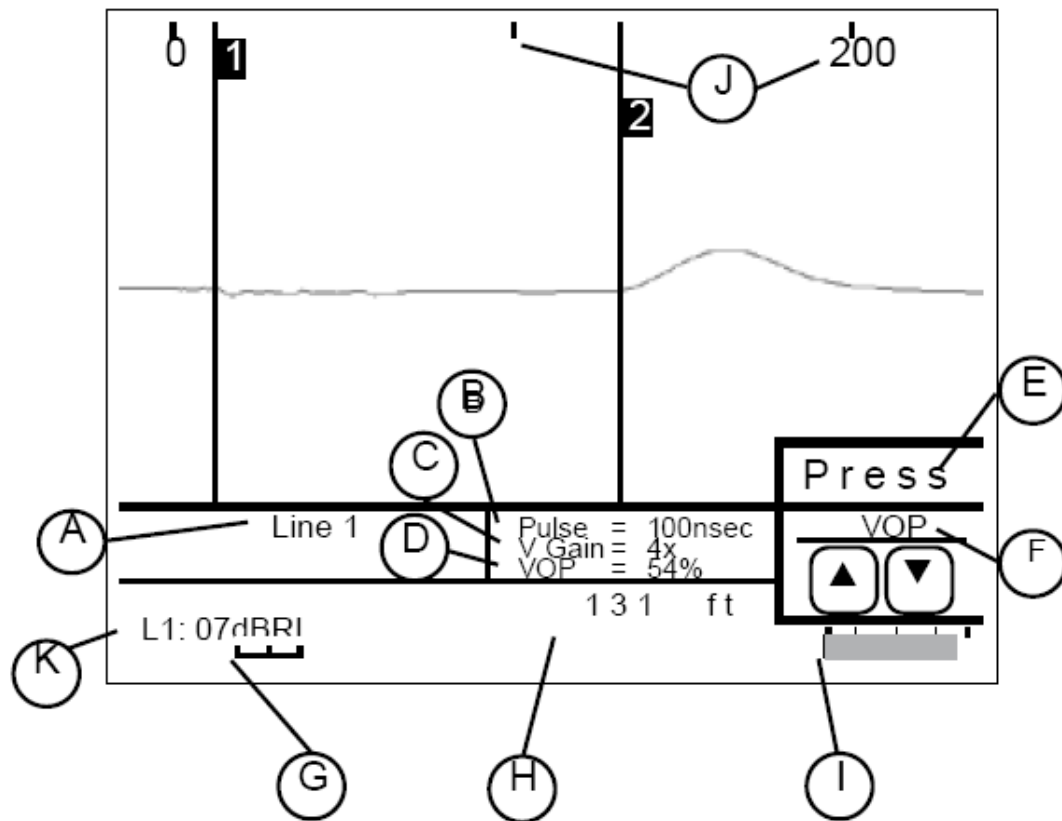
Cursors:

The first and second cursor keys move the cursors along the waveform. Use the second cursor arrows to move the second cursor to the point of interest on the waveform. Cursors must be set on the leading edges of the reflection. A pop-up menu appears when the asterisk key (*) is pressed. The unlabeled icon keys control a selection cursor for choosing the required instrument control. When the control is selected, pressing the asterisk key closes the pop-up menu and activates the control. When a control is activated, the icon keys control the function and the on-screen icons graphically represent how the keys affect the control. The icon changes depending on the type of action in the particular control.

E.2.1.2 Display

The top two-thirds of the display contains the waveform and cursors. Instrument setting and measurements are located at the bottom of the display.

Information areas on the Liquid Crystal Display (LCD) are:



A. MESSAGE CENTER

The message center displays information about the status of the instrument. Additional messages are displayed while utilizing the standard and optional waveform storage functions.

B. PULSE WIDTH

Model 1270A has selectable pulse widths for testing various lengths of cable.

C. VERTICAL GAIN

Displays the level of vertical amplitude or gain applied to the waveform.

D. VOP or V/2

The programmed velocity of propagation is displayed as a percentage of the speed of light from 30% to 99%.

E. MENU

Pop-up menu for selecting instrument controls.

F. SELECTED MENU ITEM

Displays the currently active menu selection, which is controlled by the icon keys. The on-screen icons graphically represent how the keys affect the control.

G. BALANCE (NOT Used for Co-axial Cables)

Displays the current position of the CABLE BALANCE control on a "slider" type indicator for Line 1 and Line 2.

H. DISTANCE BETWEEN CURSORS

The Model 1270A automatically calculates and displays the distance between the first and second cursors. Each time the cursor placement is changed or the VOP is adjusted, the DISTANCE BETWEEN CURSORS reading is automatically updated.

I. BATTERY LEVEL INDICATOR

A horizontal bar graph indicates the battery level. When the battery level reaches the one-quarter full scale mark, the low battery message is activated.

J. DISTANCE MARKERS

These tick marks are displayed along the top of the screen and can be in feet or meters format. These marks enable the operator to view the distance along the cable being tested.

K. FAULT SEVERITY

The Model 1270A automatically calculates the signal return loss (dBRL). The return loss and crosstalk are calculated from a comparison of the pulse amplitude at the first and second cursors. Crosstalk is only displayed in the twisted pair mode.

E.2.1.3 Pop-Up MENU**Store:**

Use the icon keys to select an available storage location and press the * key to store.

Recall:

Use the icon keys to select a storage location and press the * to recall to display.

Pulse:

Use the two icon keys for decreasing and increasing the pulse width.

Cable:

Use the two icon keys to select the cable type under test. VOP is automatically set for that cable type.

VOP or V/2: (depending on the velocity format setting chosen in the setup menu) Use the two icon keys for decreasing and increasing the velocity of propagation.

Balance:

Use the two icon keys to adjust the output balance circuit.

Filter:

Use the two icon keys for cycling through the available software filters.

Setup:

Use the * key to display the setup options menu.

Print:

Use the * key to print the on-screen waveform to a serial printer for documentation.

Mode:

Use the two icon keys for cycling through the available display modes for single, dual, or difference waveform display modes.

Test Port:

Use the two icon keys to select the active test port.

Overlay:

Use the two icon keys to adjust the trace separation in dual waveform display modes. This control is only available when two waveforms are displayed.

Tagging:

Use the * and icon keys to edit the alphanumeric label associated with a stored waveform.

Search:

Use the * key to perform an auto-search of the cable to find major faults or the end of the cable.

Press * to select.	
Store	Setup
Recall	Print
Pulse	Mode
Cable	Test Port
VOP	Overlay
Balance	Tagging
Filter	Search

E.2.1.4 Set-up the TDR Cable Tester

1. POWER: Turn power ON using the key .
2. Press the  key to open the pop-up menu list.
3. Scroll down to the “SETUP” menu and press the  key.

In the “SETUP” menu, make the following settings (Use the  key to switch between the lists of values):

- a. Horizontal Scale Units: METERS.
- b. Distance Display Format: METERS.
- c. Horizontal Reference: ON.
- d. Backlight at Start-up: OFF.
- e. Velocity Format: VOP (%).
- f. VOP Precision: 2 DIGITS.
- g. Cancel Test Lead Length: NO.
- h. Serial Printer Type: Select anything (Printer Type does not matter).

- i. Autofilter: DISABLED.
 - j. dBRL Type: TOTAL.
 - k. Language: ENGLISH.
- 4. Select EXIT and press  to close the SETUP menu.
- 5. Press  to open the pop-up menu list.
- 6. Scroll down to the “PULSE” menu and press .
- 7. Using the two   keys found just below , select the pulse based on the following criteria:
 - a. If the cable segment under test is less than 10 meters in length, select PULSE = 2 ns.
 - b. If the cable segment under test is more than 10 meters in length, select PULSE = 25 ns.
- 8. Press  to open the pop-up menu list.
- 9. Scroll down to the “CABLE” menu and press .
- 10. Using the two   keys found just below , scroll down the list of cables.
- 11. Using the  key, select the cable Comm-Scope RG -59 FOAM.
- 12. Using the two   keys found just below , scroll down the list of cables.
- 13. Select EXIT and press  to close the “Cable” menu.
- 14. Press the  key to open the pop-up menu list.
- 15. Scroll down to the “VOP” menu and press .
- 16. Using the two   keys found just below , set the VOP value as 82 %.
- 17. Press the  key to open the pop-up menu list.
- 18. Scroll down to the “Test Port” menu and press .

19. Select “coax” and press .
20. Leave the Filter, Print, Mode, Search and Tagging options untouched.
21. You may choose to use the “Store & Recall” menu later.

E.3 TEST PROCEDURE

NOTE

This test procedure is not intended to be used with multiple pieces of COAX cables connected together.

CAUTION

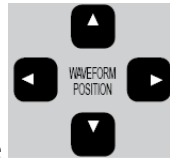
Pulse width and vertical gain are automatically adjusted for each range. Revert to the required Pulse Width and V Gain values after performing the steps provided in the following sections:

1. E.2.1.4 for Pulse Width
2. Step 1 of E.3.1 for V-Gain.

E.3.1 LCN Cables Less than 10 meters in Length

Test each length of cable, one at a time, as follows:

1. Using the steps outlined in section E.2.1.4, adjust the pulse width to 2 ns.
2. Using the   keys on the Front Panel, adjust V Gain to 1X.
3. With no COAX connected, the TDR screen must look like Figure E-1.
4. Connect the COAX to be tested to the COAX cable jack on the TDR; do not terminate the other end. The screen must look like Figure E-2.
5. Using the   Range keys, adjust the distance displayed on screen in such a way that the entire length of cable under test appears on the TDR screen.



6. Using the Waveform Position Control Keys, adjust the wave in such a way that the following criteria are met:
 - a. The base of the wave must be on the horizontal reference line.
 - b. Cursor 1 must be placed at the start of the “Transmitted Pulse”, coinciding with the “0 m” mark.
 - c. Cursor 2 must be placed at the start of the “Returned Pulse”. The screen must look like Figure E-3.
7. The distance between the two cursors is the actual length of the cable.
8. Terminate the other end of the cable (using “The selected Terminator”). The cable terminator can be attached or removed several times to verify the position of the “Returned Pulse”.
9. When the cable is not terminated and the two cursors are properly placed, the “RETURN LOSS” value is displayed at the bottom-left of the screen. The dBRL value displayed should be zero.
10. Record the dBRL value.
11. Terminate the cable (using “The selected Terminator”). The screen must look like Figure E-4.
12. Move cursor 2 to the left (towards cursor 1) by 0.6 meters.
13. A new “RETURN LOSS” value is displayed at the bottom-left of the screen. The screen must look like Figure E-5.
14. Record the “RETURN LOSS” dBRL value.
15. If the dBRL value recorded in Step 10 is a non-zero value, this dBRL value must be subtracted from the dBRL value in Step 14 to arrive at the final dBRL value.
16. Test both ends of the cable using the steps 4 through 15 of this procedure as necessary.
17. If the Final dBRL value is more than 34 dB when tested at both ends, the LCN cable is acceptable.
18. Tag all REJECTED COAX and perform the steps necessary to replace them.

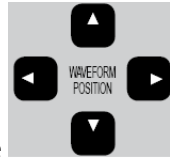
E.3.2 LCN Cables More than 10 meters in Length

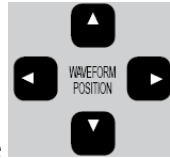
Test each length of cable, one at a time, as follows:

1. Using the steps outlined in section E.2.1.4, adjust the pulse width to 25 ns.
2. Using the  keys on the Front Panel, adjust V Gain to 1X.
3. With no COAX connected, the TDR screen must look like Figure E-1.

4. Connect the COAX to be tested to the COAX cable jack on the TDR; do not terminate the other end. The screen must look like Figure E-2.

5. Using the  Range keys, adjust the distance displayed on screen in such a way that the entire length of cable under test appears on the TDR screen.



6. Using the  Waveform Position Control Keys, adjust the wave in such a way that the following criteria are met:
 - a. The base of the wave must be on the horizontal reference line.
 - b. Cursor 1 must be placed at the start of the “Transmitted Pulse”, coinciding with the “0 m” mark.
 - c. Cursor 2 must be placed at the start of the “Returned Pulse”. The screen must look like Figure E-3.
7. The distance between the two cursors is the actual length of the cable.
8. Terminate the other end of the cable (using “The selected Terminator”). The cable terminator can be attached or removed several times to verify the position of the “Returned Pulse”.
9. When the cable is not terminated, and the two cursors are properly placed, the “RETURN LOSS” value is displayed at the bottom-left of the screen. The dBRL value displayed should be zero. The screen must look like Figure E-6.
10. Record the dBRL value.
11. Terminate the cable (using “The selected Terminator”).
12. A new “RETURN LOSS” value is displayed at the bottom-left of the screen. The screen must look like Figure E-7.
13. Record the dBRL value.
14. If the dBRL value recorded in Step 10 is a non-zero value, this dBRL value must be subtracted from the dBRL value in Step 13 to arrive at the final dBRL value.
15. Test both the ends of the cable using the steps 4 through 14 of this procedure as necessary.
16. If the final dBRL value is more than 34 dB when tested at both ends, the LCN cable is acceptable.
17. Tag all REJECTED COAX and perform the steps necessary to replace them.

E.3.3 Testing the Center of LCN Cables Longer than 10 meters

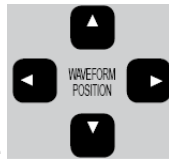
NOTE

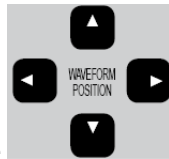
The previous tests check the ends of the COAX cable to determine if the cable is acceptable but the previous tests do not check the center of long cables. The following test provides a method for checking the center of longer cables (10 meters or more) for cable faults that would cause LCN communication failures.

Test each length of cable, one at a time, as follows:

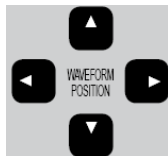
1. Using the steps outlined in section E.2.1.4, adjust the pulse width to 25 ns.
2. Using the   keys on the Front Panel, adjust V Gain to 1X.
3. With no COAX connected, the TDR screen must look like Figure E-1.
4. Connect the COAX to be tested to the COAX cable jack on the TDR; do not terminate the other end. The screen must look like Figure E-2.
5. Using the   Range keys, adjust the distance displayed on screen in such a way that the part of the entire cable length is displayed on the TDR screen.

To illustrate the statement, consider that a 150-meter cable is to be tested over its entire length. In that case, starting from “Transmitted Pulse”, you can view this cable TDR screen by selecting a range of 30 meters. Therefore, at any point in time you are able to examine 30 meters of the cable in details.



6. Using the  Waveform Position Control Keys, adjust the wave in such a way that the following criteria are met:
 - a. The Base of the wave must be on the horizontal reference line.
 - b. Cursor 1 must be placed at the start of the “Transmitted Pulse”, coinciding with the “0 m” mark.
 - c. Leaving cursor 1 at “0 m”, place cursor 2 immediately next to the “Transmitted Pulse”.
7. Using the   keys on the Front Panel, adjust V Gain to 10X.
8. Move cursor 2 away from cursor 1 until it reaches right most portion of the waveform displayed on TDR Screen. Look for variations in the waveform line that rise above or below the average level of the waveform line. Choose the highest or the lowest point of the waveform in the entire display. Do not include the “Transmitted Pulse” or the “Returned Pulse” and move cursor 2 to that location.
9. Note the return loss at that point. If that peak of the waveform has a return loss reading of 34 dBRL or greater, the reflection at that peak is acceptable. If the return loss of the peak is lower than 34dBRL, the cable is damaged at this location and must be replaced.

10. If the return loss value over the length of this cable segment is 34 dBRL or greater, then use the



Waveform Position Control Keys to shift the waveform to the left and move on to the next length of the cable segment. Adjust the wave in such a way that the following criteria are met:

- a. The Base of the wave must be on the horizontal reference line.
 - b. Leave cursor 1 at “0 m” and cursor 2 at its previous position on the TDR Screen to adjust the waveform such that cursor 2 is at the extreme left of the TDR Screen.
11. Move cursor 2 towards the right on the TDR Screen until it reaches the extreme right of the waveform displayed on the TDR screen. Look for variations in the waveform line that rise above or below the average level of the waveform line. Choose the highest or the lowest point of the waveform in the entire display. Do not include the “Transmitted Pulse” or the “Returned Pulse” and move cursor 2 to that location.
 12. Note the return loss at that point. If the peak of the waveform has a return loss reading of 34 dBRL or greater, the reflection at that peak is acceptable. If the return loss of the peak is lower than 34dBRL, the cable is damaged at this location and must be replaced.
 13. Repeat steps 10 through 12 to continue moving towards the new segment of the LCN COAX cable, until you reach “Returned Pulse”. If at any time the return loss at the peak of the waveform within the display is lower than 34dBRL, the cable is damaged and must be replaced.
 14. If the return loss at the peaks is equal to or greater than 34dBRL over the entire length of the cable, then the cable is acceptable. The test is complete.

E.4 TEST COMMENTS/CONDITIONS

1. Keep the batteries of the 1270A TDR cable tester fully charged.
2. Do not compromise (cheat) on the accuracy of the terminator. This whole procedure to ACCEPT or REJECT COAX cables depends on the delta or difference between the SELECTED TERMINATOR and the COAX UNDER TEST. Paint or identify the SELECTED terminator because it is hard to find in a normal distribution of terminators.

NOTE

Terminator specifications allow a 2% variation, that is, +/-1.5 ohms or 73.5 to 76.5 ohms. The selected terminator must be within 0.1 ohm, that is, 74.9 to 75.1 ohms. Earlier, some terminators were mistakenly made with inductive resistors that measure properly on an ohmmeter, but show as "s" rotated 90 degrees on the TDR. Do not use such terminators on the LCN or Data Hiway.

3. If absolutely necessary, you can use a "T-connector" instead of the "Splice" to connect the COAX to the terminator. However, the "T-connector" must not have the 1-ohm resistor in series or the 56 k ohm resistor to the case of the "T-connector."
4. Workmanship: The BNC connectors must be installed on the COAX with the correct tools. If the connection is made using PLIERS, CHANNEL-LOCKS, or VICE GRIPS, REJECT the COAX.
5. Check for folds or kinks in the COAX, or whether the minimum bend radius is exceeded.
6. When the COAX and TERMINATOR match each other, the returned pulse is very small or very flat on the base line. Figures E-4, or E-5, or E-7, must look almost flat at the return pulse.

FIGURES:



Figure E-1 No COAX Connected

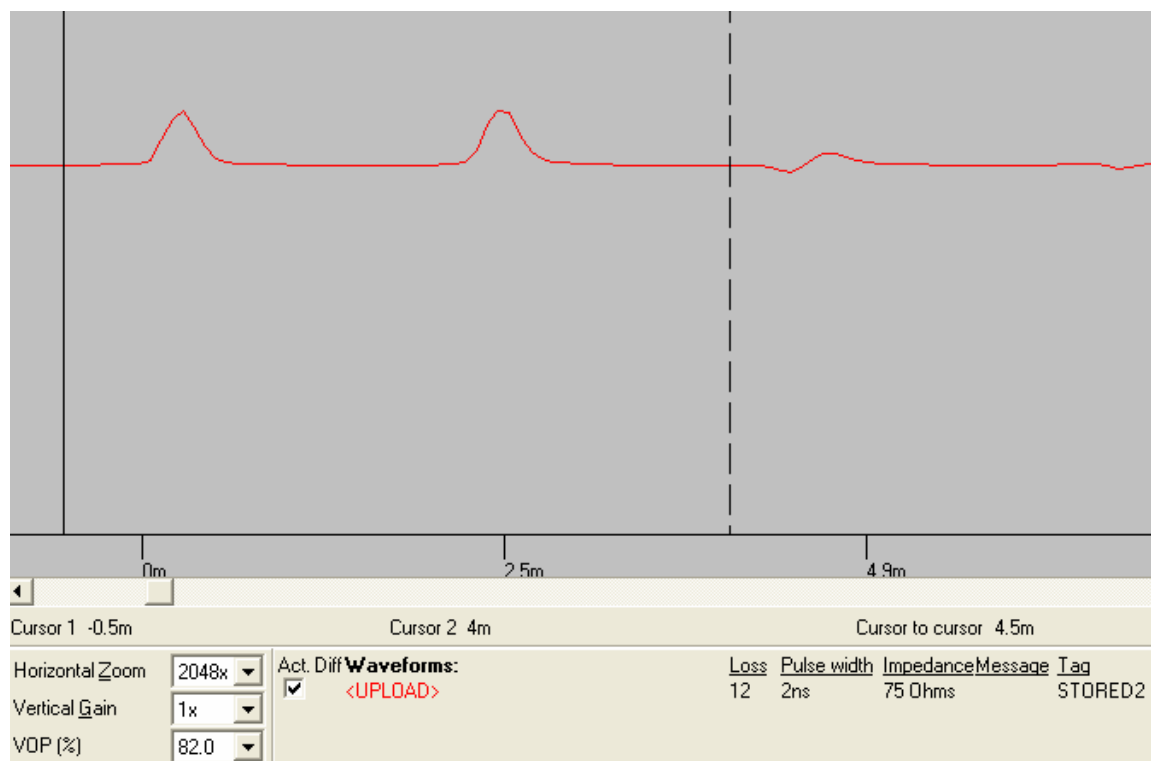


Figure E-2 COAX Connected (No Terminator)

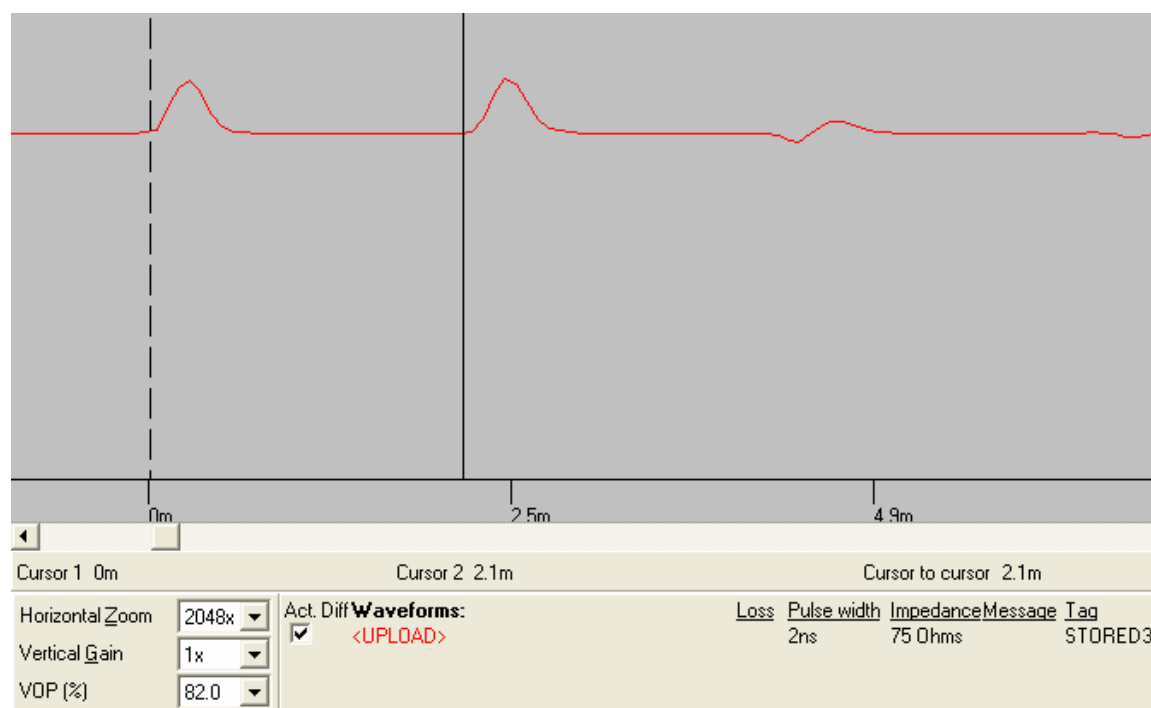


Figure E-3 COAX Connected (Range Selected, Waveform & Cursors Positioned)

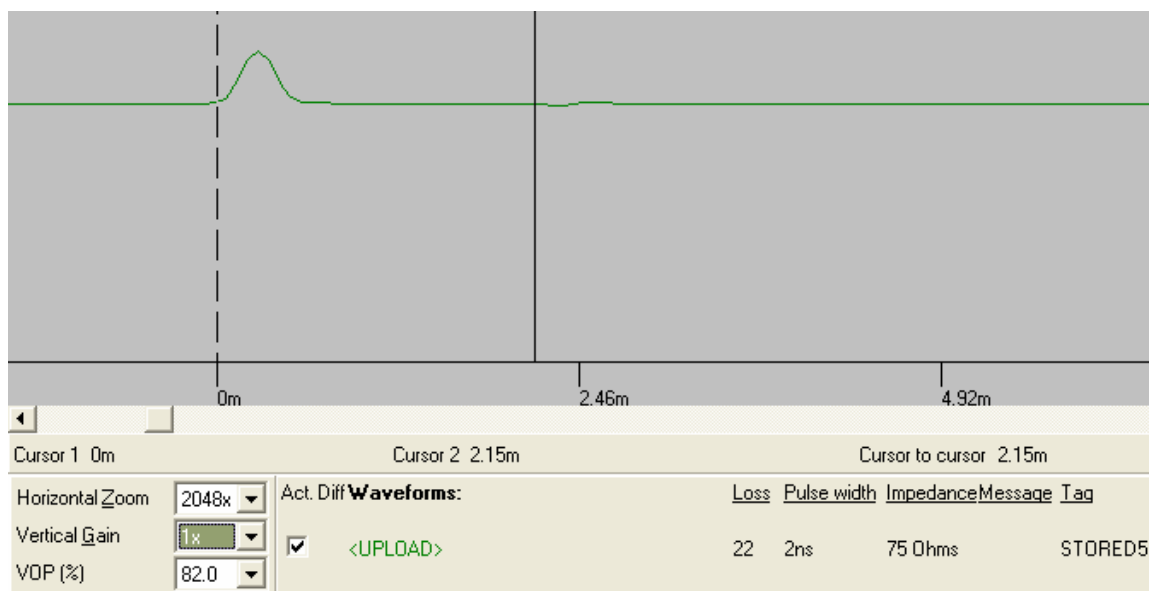


Figure E-4 COAX Terminated (2nd Cursor at Terminator)

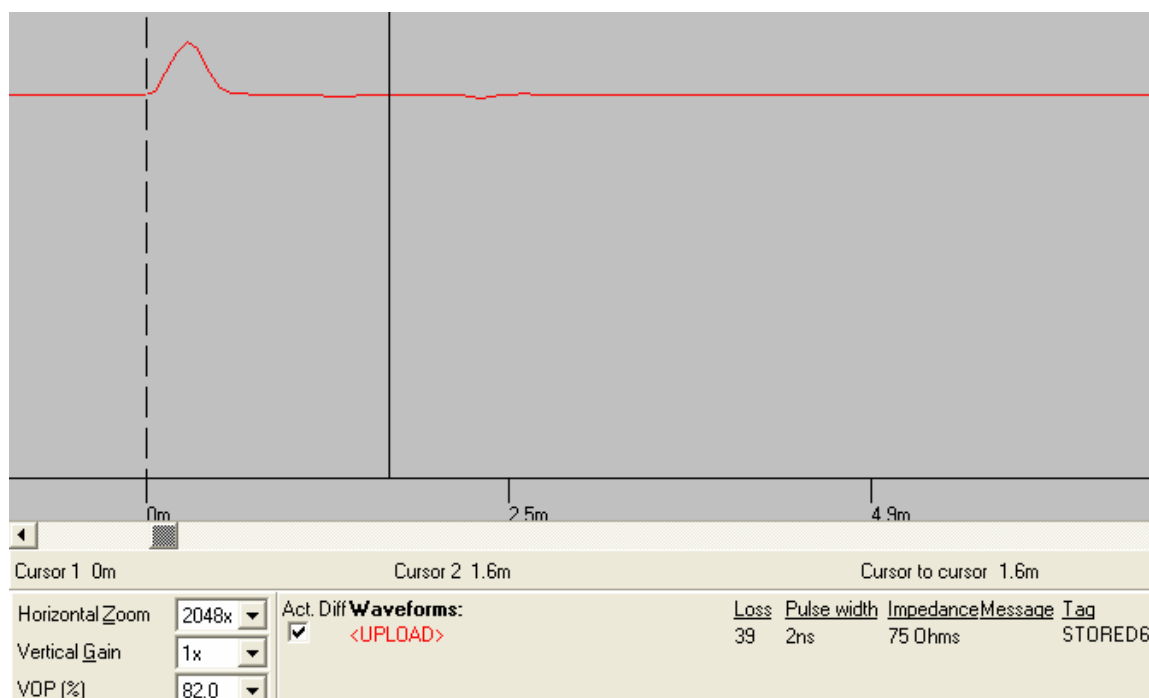


Figure E-5 COAX Terminated (2nd Cursor 0.6 Mtrs. before Terminator)

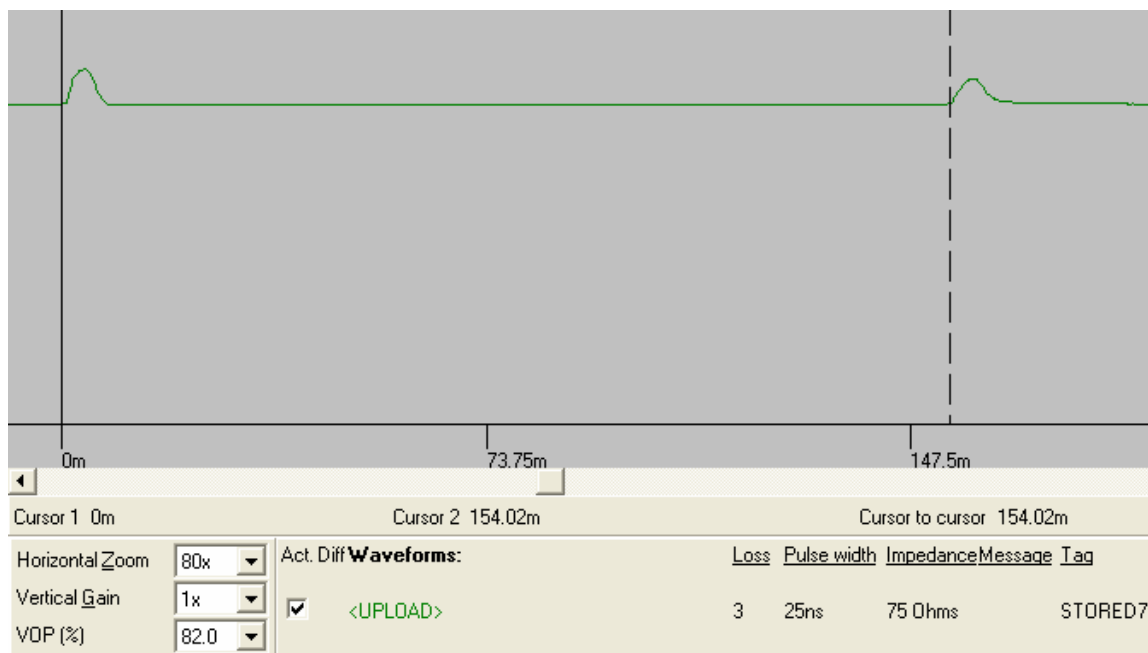


Figure E-6 COAX Connected (Range Selected, Waveform & Cursors Positioned)

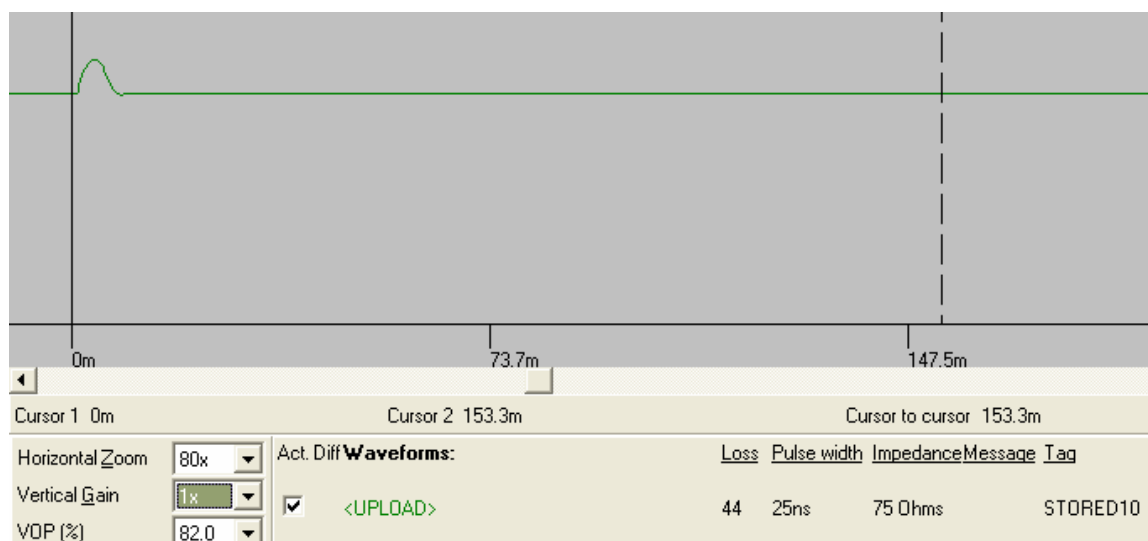


Figure E-7 COAX Terminated (2nd Cursor at Terminator)

E.5 EQUATIONS AND SAMPLE CALCULATIONS

E.5.1 Equations

1. $\text{dBRL} = 20 \log_{10} | (Z_{\text{ref}} + Z_{\text{test}}) / (Z_{\text{ref}} - Z_{\text{test}}) |$
2. $\text{dBRL} = 20 \log_{10} (V_O / V_R)$

Where, V_O is the amplitude of the transmitted pulse, and

V_R is the amplitude of the reflected pulse.

E.5.2 Sample Calculations when the Terminator Varies

	(COAX)	(TERMINATOR)	(PULSE)			RETURN
	Z ref	Z test	polarity	p	%	LOSS dB
NG	75	INF	+	1.0000	100.000	0.0000
NG	75	1.0 Meg	+	0.9998	99.985	0.0013
NG	75	1.0 k	+	0.8604	86.046	1.3053
NG	75	225.0	+	0.5000	50.000	6.0205
NG	75	100.0	+	0.1428	14.285	16.9019
OKAY	75	78.12	+	0.0203	2.037	33.8175
OKAY	75	75.10	+	0.0006	0.066	63.5276
OKAY	75	75.00	None	0.0000	0.000	INF
OKAY	75	74.90	-	0.0006	0.066	63.5160
OKAY	75	72.00	-	0.0204	2.040	33.8036
NG	75	50.0	-	0.2000	20.000	13.9794
NG	75	25.0	-	0.5000	50.000	6.0205
NG	75	0.0	-	1.0000	100.000	0.0000

E.5.3 Sample Calculations when the Coax Varies

	(COAX) Z ref	(TERMINATOR) Z test	(PULSE) polarity	p	%	RETURN LOSS dB
OKAY	78.12	75	-	0.0203	2.037	33.8175
OKAY	75.10	75	-	0.0006	0.066	63.5276
OKAY	75.00	75	None	0.0000	0.000	INF
OKAY	74.90	75	+	0.0006	0.066	63.5160

Honeywell

Honeywell Process Solutions

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